PolymerBenchmark2026｜四大源码图谱与函数接口详解

基于上传 ZIP 的逐文件静态 AST 审查｜2026-11-05｜范围：全部 Python 文件、函数/类、静态调用关系、关键张量流、训练数据流及 3D 扩展映射

可信度说明：文件、函数名、定义行号、源码表达式和静态调用来自上传文件。动态 dispatch、第三方依赖、运行时形状和分支不能通过静态 AST 完全确认。

1. 全项目数据流图

CSV/JSONL -> Datasets/Dataset_construction_scripts/* -> PSMILES / wPSMILES
PSMILES -> RDKit descriptors -> Models/RDKit_RF/train_rf.py -> RF -> metrics
PSMILES -> polyBERT embedding dictionary -> Models/polyBERT/train_pBERT.py -> FFN -> metrics
PSMILES -> polymer_periodic_graph/Polymer_Model/train.py -> data/utils.py -> data/data.py
-> features/featurization.py -> models/mpn.py -> models/model.py
-> train/train.py -> train/evaluate.py -> checkpoint -> train/make_predictions.py
wPSMILES -> polymer_chemprop shell entry -> external dependency (not present as source)
Aligned rows -> create_scaling_splits.py -> train_*_scaling.py -> summarize_scaling_results.py

2. Periodic Graph 的实际张量流（依据 mpn.py ）

BatchMolGraph.get_components()
f_atoms: [N_atom+padding, F_atom] ｜ f_bonds: [N_dirbond+padding, F_bond]
a2b: [N_atom+padding, max_degree] ｜b2a: [N_dirbond+padding]
b2revb: [N_dirbond+padding] ｜a_scope: list[(start, n_atoms)]
W_i(f_bonds): [N_dirbond+padding, H] (when atom_messages=False)
depth-1 iterations: index_select_ND -> neighbor sum -> reverse-bond subtraction -> W_h -> activation/dropout
index_select_ND(message,a2b): [N_atom+padding,max_degree,H]
concat(f_atoms, summed message): [N_atom+padding,F_atom+H]
W_o -> atom_hiddens: [N_atom+padding,H]
readout by a_scope (mean/sum/norm) -> mol_vecs: [B,H]
MoleculeModel FFN -> predictions: [B,T] for T regression tasks (subject to configuration)

注意：N_atom/N_dirbond 包含实现使用的 padding 索引；F_bond 的实际定义依赖 atom_messages 和 featurization ；H 来自 args.hidden_size，T 来自 args.num_tasks。请勿把示意变量误当作固定数值。

3. 模型内部源码证据与公式

Models/polymer_periodic_graph/Polymer_Model/chemprop/features/featurization.py

函数	源码行号	输入参数	返回标注	调用/操作
get_atom_fdim	L73–80	overwrite_default_atom	int	
get_bond_fdim	L127–142	atom_messages, overwrite_default_bond, overwrite_default_atom	int	get_atom_fdim
atom_features	L189–210	atom,	List[Union[bool,	atom.GetAtomicN

		functional_groups	int, float]]	um、onek_encoding_unk、atom.GetMass、int、atom.GetIsAromatic、atom.GetHybridization、atom.GetTotalNumHs、atom.GetFormalCharge、atom.GetChiralTag、atom.GetTotalDegree
bond_features	L228–249	bond	List[Union[bool, int, float]]	bond.GetBondType、onek_encoding_unk、int、list、bond.GetIsConjugated、bond.IsInRing、bond.GetStereo、range
MolGraph.__init__	L341–539	self, mol, atom_features_extra, bond_features_extra, overwrite_default_atom_features, overwrite_default_bond_features	未标注；无显式 return	is_reaction、is_explicit_h、reaction_mode、Chem.AddHs、Chem.MolToMolBlock、output_mol_array.split、mol.GetNumAtoms、mol.GetAtoms、np.array、type、atom.GetNeighbors
BatchMolGraph.__init__	L556–603	self, mol_graphs	未标注；无显式 return	get_atom_fdim、get_bond_fdim、max、torch.FloatTensor、torch.LongTensor、f_atoms.extend、f_bonds.extend、range、self.a_scope.append、self.b_scope.append、a2b.append
BatchMolGraph.get_components	L605–633	self, atom_messages	Tuple[torch.FloatTensor, torch.FloatTensor, torch.LongTensor, torch.LongTensor, torch.LongTensor,	get_bond_fdim

			List[Tuple[int, int]], List[Tuple[int, int]]]	
BatchMolGraph.get_b2b	L635–647	self	torch.LongTensor	(b2b != self.b2revb.unsqueeze(1).repeat(1, b2b.size(1))).long、self.b2revb.unsqueeze(1).repeat、b2b.size、self.b2revb.unsqueeze
BatchMolGraph.get_a2a	L649–662	self	torch.LongTensor	

Models/polymer_periodic_graph/Polymer_Model/chemprop/models/mpn.py

函数	源码行号	输入参数	返回标注	调用/操作
MPNEncoder.__init__	L17–64	self, args, atom_fdim, bond_fdim	未标注；无显式 return	super(MPNEncoder, self).__init__、nn.Dropout、get_activation_function、nn.Parameter、nn.Linear、torch.zeros、super
MPNEncoder.forward	L66–150	self, mol_graph, atom_descriptors_batch	torch.FloatTensor	mol_graph.get_components、self.act_func、range、index_select_ND、nei_a_message.sum、torch.cat、self.dropout_layer、enumerate、torch.stack、torch.from_numpy(np.concatenate(atom_descriptors_batch, axis=0)).float().to、f_atoms.to
MPN.forward	L187–266	self, batch, features_batch, atom_descriptors_batch, atom_features_batch, bond_features_batch	torch.FloatTensor	reduce、type、torch.from_numpy(np.stack(features_batch)).float().to、torch.cat、len、NotImplementedError、enc、features_batch.view、range、mol2graph、torch.from_numpy(np.stack(features_batch)).float

Models/polymer_periodic_graph/Polymer_Model/chemprop/models/model.py

函数	源码行号	输入参数	返回标注	调用/操作
MoleculeModel._	L17–39	self, args	未标注；无显式	super(MoleculeMo

_init__			return	del, self).__init__、self.create_encoder、self.create_ffn、initialize_weights、nn.Sigmoid、nn.Softmax、super
MoleculeModel.create_encoder	L41–55	self, args	None	MPN、list(self.encoder.encoder.children())[0].parameters、self.encoder.parameters、list、self.encoder.encoder.children
MoleculeModel.create_ffn	L57–121	self, args	None	nn.Dropout、get_activation_function、nn.Sequential、range、ffn.extend、ffn.append、nn.Linear、nn.Softplus、nn_exp、list、super(nn_exp, self).__init__
MoleculeModel.forward	L152–185	self, batch, features_batch, atom_descriptors_batch, atom_features_batch, bond_features_batch	torch.FloatTensor	self.ffn、self.encoder、self.sigmoid、output.reshape、self.multiclass_softmax、output.size

有向键模式的核心聚合（mpn.py）：message = a_message[b2a] - message[b2revb]，排除反向键的即时回传。然后 message = act(W_i(input) + W_h(message))；Readout 对每个 a_scope 的原子向量执行 mean/sum/norm。

4. CSV/JSONL → Dataset → Batch → Forward → Loss → Checkpoint → Prediction

1. 原始数据解析

文件：Datasets/Dataset_construction_scripts/；主要对象：process_Vipea_data.py / OMers_convert_jsonl.py；输出/作用：结构、标签、元数据转换

2. 标准化结构

文件：Datasets/Dataset_construction_scripts/polymer_conversions.py；主要对象：函数级转换逻辑；输出/作用：PSMILES/wPSMILES

3. 数据加载

文件：chemprop/data/utils.py；主要对象：get_data / get_data_from_smiles（以源码为准）；输出/作用：MoleculeDataset

4. Dataset/Batch

文件：chemprop/data/data.py；主要对象：MoleculeDatapoint / MoleculeDataset / MoleculeDataLoader；输出/作用：MoleculeDatapoint 列表、BatchMolGraph

5. 图构建

文件：chemprop/features/featurization.py；主要对象：MolGraph / BatchMolGraph；输出/作用：f_atoms,f_bonds,a2b,b2a,b2revb

6. Forward

文件：chemprop/models/model.py；主要对象：MoleculeModel.forward；输出/作用：预测张量

7. Message Passing

文件：chemprop/models/mpn.py；主要对象：MPNEncoder.forward；输出/作用：[B,H]

8. Loss/Optimization

文件：chemprop/train/train.py；主要对象：train；输出/作用：loss/backward/optimizer step

9. Validation

文件：chemprop/train/evaluate.py；主要对象：evaluate；输出/作用：验证分数

10. Checkpoint

文件：chemprop/train/run_training.py；主要对象：run_training；输出/作用：保存最优模型

11. Prediction

文件：chemprop/train/make_predictions.py；主要对象：make_predictions；输出/作用：测试集预测结果

5. 逐文件完整函索引（含用系）

扫描 Python 文件 72 个，成功解析 72 个。每个函数/方法显示定义行号、参数、返回类型注解、静态识别的直接调用。调用列表是 AST 法用，不代表被行正行；self.*、第三方库、函数别名可能无法静态消歧。

Datasets/Dataset_construction_scripts/OMers_convert_jsonl.py

函数/类	行号	参数	返回	静态调用（前 14 项）
construct_alternating_wPSMILES_from_row	32–37	OMers_row	未标注；含 return	construct_alternating_wPSMILES
construct_random_wPSMILES_from_row	40–45	OMers_row	未标注；含 return	construct_random_wPSMILES

Datasets/Dataset_construction_scripts/PSMILES_to_RDKit_descriptors.py

函数/类	行号	参数	返回	静态调用（前 14 项）
is_csv_like	22–24	path	bool	path.name.lower().endswith, path.name.lower
process_file	27–67	input_path,	未标注；含 return	print, pd.read_csv,

		output_path, cache		cache.descriptor_frame, pd.concat, output_path.parent.mkdir, result.to_csv, len
generate_rdkit_descriptors	70–132	cache_path	未标注；含 return	print, default_cache_path, psmiles_root.exists, DescriptorCache, psmiles_root.rglob, p.is_file, input_path.relative_to, output_path.exists, process_file, len

Datasets/Dataset_construction_scripts/PSMILES_to_pBERT.py

函数/类	行号	参数	返回	静态调用（前 14 项）
is_csv_like	10–18	path	bool	path.name.lower, name.endswith

Datasets/Dataset_construction_scripts/PSMILES_to_wPSMILES.py

函数/类	行号	参数	返回	静态调用（前 14 项）
is_csv_like	29–37	path	bool	path.name.lower, name.endswith
process_file	39–82	in_path, in_root, out_root	None	in_path.relative_to, out_path.parent.mkdir, print, pd.read_csv, df.drop_duplicates('smiles').copy, df_out['smiles'].apply, df_out.to_csv, ValueError, df.drop_duplicates
main	84–132		int	argparse.ArgumentParser, ap.add_argument, ap.parse_args, args.input_dir.resolve, args.output_dir.resolve, print, in_root.exists, in_root.is_dir, in_root.rglob, process_file, p.is_file, is_csv_like, args.pattern.lower, p.name.lower

Datasets/Dataset_construction_scripts/combine_alternating_homopolymer.py

函数/类	行号	参数	返回	静态调用（前 14 项）
main	8–32		未标注；无显式 return	argparse.ArgumentParser,

				ap.add_argument, ap.parse_args, args.input_dir.resolve, str, glob.glob, pd.concat, re.match, combined_df.to_csv, pd.read_csv, m.groups, Path, two_dirs.split, csv.split('/')[-1].split

Datasets/Dataset_construction_scripts/convert_web_datasets.py

函数/类	行号	参数	返回	静态调用（前 14 项）
convert_web_datasets	24–92	psmiles_root, wpsmiles_root, datasets, strict_missing	未标注；含 return	Path(psmiles_root or script_dir / '../../Datasets/PSMILES').resolve, Path(wpsmiles_root or script_dir / '../../Datasets/wPSMILES').resolve, print, Path, input_dir.exists, subprocess.run, str
parse_args	95–105		未标注；含 return	argparse.ArgumentParser, parser.add_argument, parser.parse_args, list

Datasets/Dataset_construction_scripts/create_PSMILES_pBERT_dictionary.py

函数/类	行号	参数	返回	静态调用（前 14 项）
parse_args	15–24		argparse.Namespace	argparse.ArgumentParser, parser.add_argument, parser.parse_args, Path, os.environ.get
is_csv_like	27–28	path	bool	path.name.lower().endswith, path.name.lower
main	31–78		None	parse_args, args.model_path.resolve, dictionary_path.exists, SentenceTransformer, model_path.is_dir, SystemExit, print, str, save_dictionary, dictionary_path.open, pickle.load, input_root.rglob, pickle.dump, pd.read_csv(dataset).smiles.tolist

Datasets/Dataset_construction_scripts/create_scaling_splits.py

函数/类	行号	参数	返回	静态调用（前 14 项）
repo_root	32–33		Path	Path(__file__).resolve, Path
parse_args	36–95		argparse.Namespace	argparse.ArgumentParser, parser.add_argument, parser.parse_args, repo_root
target_columns	98–101	df, representation	list[str]	—
require_single_target	104–110	df, representation	str	target_columns, len, ValueError
dataset_path	113–114	root, representation, dataset, prop, filename	Path	—
load_and_validate_inputs	117–164	args	tuple[dict[str, pd.DataFrame], str]	dataset_path, pd.read_csv, require_single_target, pd.to_numeric(psmiles[target_col], errors='raise').to_numpy, psmiles_path.exists, FileNotFoundError, ValueError, np.isfinite(canonical_y).all, pd.to_numeric(df[rep_target], errors='raise').to_numpy, ImportError, pd.to_numeric, path.exists, len, np.allclose
build_rdkit_frame	167–188	psmiles, target_col, cache_path	pd.DataFrame	print, pd.concat, psmiles['smiles'].apply, sanitize_descriptors, ValueError, pd.DataFrame, int, DescriptorCache, cache.descriptor_frame, psmiles[[target_col]].reset_index, len, descriptor_rows.items
write_csv	191–195	df, path, overwrite	None	path.parent.mkdir, df.to_csv, path.exists
split_indices	198–214	n_rows, fold, seed, val_frac, test_frac	tuple[np.ndarray, np.ndarray, np.ndarray]	np.random.default_rng, np.arange, rng.shuffle, int, ValueError, round
indices_frame	217–227	indices, split, psmiles, target_col	pd.DataFrame	np.asarray, pd.DataFrame, list, np.arange, psmiles.iloc[idx]['smiles'].to_numpy, psmiles.iloc[idx]

				[target_col].to_numpy, len
output_base	230–231	args	Path	Path
load_published_indices	234–297	args, psmiles_path, n_rows, target_col	tuple[dict, list[tuple[np.ndarray, np.ndarray, np.ndarray]]] | No	json.loads, expected.items, enumerate, manifest_path.exists, manifest_path.read_text, ValueError, sha256_file, len, validate_partition, folds.append, manifest.get, np.isclose, set, fold_record.get
main	300–424		None	parse_args, args.datasets_root.resolve, args.output_root.resolve, args.indices_root.resolve, sorted, load_and_validate_inputs, len, max, dataset_path, load_published_indices, output_base, range, base.mkdir, print

Datasets/Dataset_construction_scripts/create_task4_shared_splits.py

函数/类	行号	参数	返回	静态调用（前 14 项）
sha256	30–35	path	str	hashlib.sha256, digest.hexdigest, path.open, iter, digest.update, handle.read
require_columns	38–41	frame, columns, path	None	set, ValueError, sorted
read_vipea	44–64	path, architecture	pd.DataFrame	pd.read_csv, require_columns, pd.DataFrame, path.is_file, FileNotFoundError, result['psmiles'].isna().any, ValueError, frame['smiles'].map, pd.to_numeric, np.arange, np.isfinite(result[TARGET]).all, len, result['psmiles'].isna, np.isfinite
read_polyverse	67–82	psmiles_path, wpsmiles_path	pd.DataFrame	pd.read_csv, require_columns, pd.to_numeric(psmiles['electron_aff'], errors='raise').to_numpy,

				pd.to_numeric(wpsmiles['electron_aff'], errors='raise').to_numpy, pd.DataFrame, len, ValueError, np.array_equal, pd.to_numeric, np.arange
canonical_training_frame	85–105	vipea_path, polyverse_psmiles, polyverse_wpsmiles	pd.DataFrame	pd.concat([read_vipea(vipea_path, 'alternating'), read_polyverse(polyverse_psmiles, polyverse_wpsmiles)], ignore_index=True).sort_values, source.groupby('psmiles', sort=False, as_index=False).agg, canonical['architecture'].value_counts().to_dict, canonical.insert, ValueError, np.arange, pd.concat, source.groupby, canonical['architecture'].value_counts, len, read_vipea, read_polyverse, ';'.join, map
holdout_frame	108–120	vipea_root	pd.DataFrame	pd.concat, holdout.insert, holdout['architecture'].value_counts().to_dict, read_vipea, frame['wpsmiles'].map, pd.DataFrame, frames.append, np.arange, ValueError, parsed.tolist, len, holdout['architecture'].value_counts
assign_folds	123–139	train, num_folds, seed	np.ndarray	np.empty, train.groupby, ValueError, len, np.random.default_rng, rng.shuffle, np.zeros, sorted, set, int, rows.index[positions].to_numpy, rows.groupby('psmiles', sort=True).indices.items, sum, np.argmin

index_frame	142–146	frame, split	pd.DataFrame	pd.DataFrame, np.arange, frame['row_index'].to_numpy, frame['psmiles'].to_numpy, frame[TARGET].to_numpy, len
embedding_frame	149–160	frame, dictionary_path	pd.DataFrame	np.asarray, pd.DataFrame, frame[TARGET].to_numpy, dictionary_path.open, pickle.load, ValueError, len
write_representations	163–173	frame, split_dir, split, cache, dictionary_path	None	index_frame(frame, split).to_csv, frame[['psmiles', TARGET]].rename(columns={'psmiles': 'smiles'}).to_csv, frame[['wpsmiles', TARGET]].rename(columns={'wpsmiles': 'smiles'}).to_csv, cache.descriptor_frame, pd.concat([features, frame[[TARGET]].reset_index(drop=True)], axis=1).to_csv, embedding_frame(frame, dictionary_path).to_csv, index_frame, frame[['psmiles', TARGET]].rename, frame[['wpsmiles', TARGET]].rename, ValueError, pd.concat, embedding_frame, frame[[TARGET]].reset_index
create_shared_splits	176–223	output_dir, vipea_root, polyverse_psmiles, polyverse_wpsmiles, num_folds, seed, dictionary_path, with_rdkit, overwrite	None	output_dir.mkdir, canonical_training_frame, holdout_frame, train.to_csv, holdout.to_csv, holdout[['wpsmiles', TARGET]].rename(columns={'wpsmiles': 'smiles'}).to_csv, pd.DataFrame({'psmiles': pd.unique(pd.concat([holdout['psmiles_a'],

				holdout['psmiles_b']], ignore_index=True))}).to_csv, train.copy, assign_folds, (output_dir / 'manifest.json').write_text, ValueError, output_dir.exists, any, shutil.rmtree
main	226–238		None	argparse.ArgumentParser, parser.add_argument, parser.parse_args, create_shared_splits

Datasets/Dataset_construction_scripts/manage_scaling_indices.py

函数/类	行号	参数	返回	静态调用（前 14 项）
sha256_file	21–26	path	str	hashlib.sha256, digest.hexdigest, path.open, iter, digest.update, handle.read
read_row_indices	29–34	path	np.ndarray	pd.read_csv, frame['row_index'].to_numpy, len, ValueError, np.unique
validate_partition	37–50	train_pool, val, test, n_rows, context	None	np.concatenate, len, ValueError, np.array_equal, np.unique, np.sort, np.arange
write_deterministic_npz	53–67	path	None	path.parent.mkdir, zipfile.ZipFile, sorted, io.BytesIO, np.lib.format.write_array, zipfile.ZipInfo, archive.writestr, np.asarray, buffer.getvalue
experiment_relative_path	70–76	manifest	Path	Path
export_experiment	79–172	split_manifest_path, datasets_root, output_root, overwrite	Path	json.loads, experiment_relative_path, pd.read_csv, range, output_base.mkdir, output_manifest_path.write_text, split_manifest_path.read_text, canonical_path.is_file, FileNotFoundError, list, ValueError, len,

				output_manifest_path.exists, FileExistsError
verify_experiment	175–231	manifest_path, datasets_root, allow_missing_sources	None	json.loads, canonical_path.exists, enumerate, manifest_path.read_text, manifest.get, ValueError, pd.read_csv, len, validate_partition, sha256_file, list, FileNotFoundError, fold_record.get, np.load
discover_manifests	234–235	root	list[Path]	sorted, root.rglob
parse_args	238–252		argparse.Namespace	argparse.ArgumentParser, parser.add_subparsers, subparsers.add_parser, export.add_argument, verify.add_argument, parser.parse_args
main	255–280		None	parse_args, discover_manifests, print, args.split_root.resolve, FileNotFoundError, export_experiment, args.indices_root.resolve, verify_experiment, args.datasets_root.resolve, args.output_root.resolve, len

Datasets/Dataset_construction_scripts/polymer_conversions.py

函数/类	行号	参数	返回	静态调用（前 14 项）
replace_astrix	22–42	input_string, unit_number	str	input_string.split, len, print
PSMILES_to_wPSMILES	45–58	input_string	str	replace_astrix
construct_alternating_wPSMILES	61–107	copolymer_ratio, smiles_list	str	smiles_list.split, float, replace_astrix, len, range, ''.join, copolymer_ratio.split, enumerate, '.'.join, '|'.join, map, suffix_list.append
construct_random_wPSMILES	110–134	copolymer_ratio, smiles_list	str	smiles_list.split, ''.join, float, replace_astrix, len, range,

				copolymer_ratio.split, enumerate, suffix_list.append, '.'.join, '|'.join, map
wPSMILES_to_PSMILES_homopolymer_simple	141–155	wpsmiles	str	molecule.replace('[*:1]', '*').replace, wpsmiles.split, molecule.replace
join_by_pairs	158–229	mols_or_smiles, pairs, bond_orders	未标注；含 return	isinstance, combo.GetAtoms, Chem.RWMol, sorted, rw.GetMol, Chem.SanitizeMol, Chem.CombineMols, bond_orders.get, rw.AddBond, to_delete.extend, set, rw.RemoveAtom, Chem.MolFromSmiles, mols.extend
wPSMILES_to_PSMILES_alternating	232–267	input_smiles	str	input_smiles.split, join_by_pairs, Chem.MolToSmiles, re.compile, pattern.sub, len, ValueError, range
is_homopolymer	274–285	wpsmiles	bool	wpsmiles.split
get_architecture	288–310	wpsmiles	str	is_homopolymer, wpsmiles.split

Datasets/Dataset_construction_scripts/process_Vipea_data.py

函数/类	行号	参数	返回	静态调用（前 14 项）
process_vipea_data	11–58	script_dir, datasets_root	None	Path, pd.read_csv, wpsmiles_data.reset_index().merge, polymer_information.reset_index, wpsmiles_data.reset_index, df[df.poly_type == architecture].reset_index, wpsmiles_path.parent.mkdir, df_temp[['smiles', quantity]].to_csv, df_temp.smiles.apply, psmiles_path.parent.mkdir

Datasets/Dataset_construction_scripts/rdkit_descriptor_cache.py

函数/类	行号	参数	返回	静态调用（前 14 项）
calculate_descriptors	34–39	psmiles	dict[str, float]	Chem.MolFromSmiles, CalcMolDescriptors
descriptor_columns	42–45		list[str]	calculate_descriptors
sanitize_descript	48–54	features, columns	pd.DataFrame	features.reindex,

ors				features.apply, features.replace, descriptor_columns
default_cache_path	57–59	datasets_root	Path	—
_json_value	62–64	value	float | None	float, math.isfinite
DescriptorCache	67–160	class	类实例	—
DescriptorCache.__init__	70–75	self, path	未标注；无显式 return	Path, self.path.parent.mkdir, sqlite3.connect, descriptor_columns, self._initialize_or_validate
DescriptorCache.__enter__	77–78	self	'DescriptorCache'	—
DescriptorCache.__exit__	80–81	self	None	self.close
DescriptorCache.close	83–84	self	None	self.connection.close
DescriptorCache.metadata	86–93	self	dict[str, object]	list
DescriptorCache._initialize_or_validate	95–120	self	None	self.connection.execute, self.connection.execute("SELECT value FROM metadata WHERE key = 'provenance'").fetchone, self.metadata, json.loads, self.connection.commit, ValueError, json.dumps
DescriptorCache.descriptor_rows	122–154	self, psmiles	list[dict[str, float]]	list, range, calculated.items, dict.fromkeys, len, ','.join, self.connection.execute(f'SELECT psmiles, payload FROM descriptors WHERE psmiles IN ({placeholders})', batch).fetchall, cached.update, calculate_descriptors, self.connection.executemany, self.connection.commit, records.append, self.connection.execute, json.loads
DescriptorCache.descriptor_frame	156–160	self, psmiles	tuple[pd.DataFrame, list[int]]	self.descriptor_rows, sanitize_descriptors, enumerate, pd.DataFrame

Datasets/Dataset_construction_scripts/task4_architecture_transfer.py

函数/类	行号	参数	返回	静态调用（前 14 项）
parse_wpsmiles	28–42	wpsmiles	tuple[str, str, float, float]	wpsmiles.split, parts[0].split, float, ValueError, any, MAP_PATTERN.sub, len, abs, sum
weighted_prediction	45–49	prediction_a, prediction_b, fraction_a, fraction_b	float	ValueError, abs
build_task4	52–81	train_path, info_path, wpsmiles_path, output_dir	None	pd.read_csv, pd.DataFrame, info.join, merged[merged['poly_type'].isin(['random', 'block'])].copy, holdout['poly_chemprop_input'].map, output_dir.mkdir, train_output.to_csv, holdout.rename(columns={'poly_id': 'polymer_id', 'poly_type': 'architecture', 'poly_chemprop_input': 'wpsmiles', 'EA (eV)': 'electron_affinity'}) [['polymer_id', 'architecture', 'wpsmiles', 'psmiles_a', 'psmiles_b', 'molar_fraction_a', 'molar_fraction_b', 'electron_affinity']].to_csv, ValueError, len, parsed.tolist, (holdout['fracA'].astype(float).round(12) == holdout['molar_fraction_a'].round(12)).all, (holdout['fracB'].astype(float).round(12) == holdout['molar_fraction_b'].round(12)).all, list
main	84–91		None	argparse.ArgumentParser, parser.add_argument, parser.parse_args, build_task4

Datasets/Dataset_construction_scripts/wPSMILES_to_PSMILES.py

函数/类	行号	参数	返回	静态调用（前 14 项）
wPSMILES_to_PSMILES_homopolymer	16–26	dataset	未标注；含 return	dataset.poly_chemprop_input.str.replace, dataset.poly_chemprop_input.str.split

Datasets/task4_architecture_transfer/evaluate_rdkit_ood.py

函数/类	行号	参数	返回	静态调用（前 14 项）
metrics	21–29	frame	dict[str, float | int]	float, np.mean, int, np.square, len, np.sqrt, np.abs
main	32–72		None	argparse.ArgumentParser, parser.add_argument, parser.parse_args, pd.read_csv, pd.unique, args.output_root.mkdir, sorted, (args.output_root / 'ood_metrics_all_folds.json').write_text, pd.concat, DescriptorCache, cache.descriptor_frame, ValueError, args.result_root.glob, joblib.load

Datasets/task4_architecture_transfer/summarize_weighted_predictions.py

函数/类	行号	参数	返回	静态调用（前 14 项）
weighted_prediction	14–20	prediction_a, prediction_b, fraction_a, fraction_b	float	ValueError, np.isclose
metrics	23–31	frame	dict[str, float | int]	float, np.mean, int, np.square, len, np.sqrt, np.abs
main	34–72		None	argparse.ArgumentParser, parser.add_argument, parser.parse_args, pd.read_csv, predictions['psmiles'].duplicated().any, holdout['psmiles_a'].map, holdout['psmiles_b'].map, holdout[['prediction_a', 'prediction_b']].isna().any().any, holdout.apply, args.output.parent.

				mkdir, holdout.to_csv, print, ValueError, predictions.set_index

Models/OpenAI/EA_predictions/homopolymer_EA_4o_neg.py

函数/类	行号	参数	返回	静态调用（前 14 项）
construct_prompt	13–25	PSMILES, phyiscal_property, units_requested, use_common_name	未标注；含 return	—

Models/OpenAI/EA_predictions/homopolymer_EA_4o_pos.py

函数/类	行号	参数	返回	静态调用（前 14 项）
construct_prompt	14–26	PSMILES, phyiscal_property, units_requested, use_common_name	未标注；含 return	—

Models/OpenAI/EA_predictions/homopolymer_EA_o3_neg.py

函数/类	行号	参数	返回	静态调用（前 14 项）
construct_prompt	14–26	PSMILES, phyiscal_property, units_requested, use_common_name	未标注；含 return	—

Models/OpenAI/EA_predictions/homopolymer_EA_o3_pos.py

函数/类	行号	参数	返回	静态调用（前 14 项）
construct_prompt	14–26	PSMILES, phyiscal_property, units_requested, use_common_name	未标注；含 return	—

Models/OpenAI/IP_predictions/homopolymer_IP_4o_neg.py

函数/类	行号	参数	返回	静态调用（前 14 项）
construct_prompt	14–26	PSMILES, phyiscal_property, units_requested, use_common_name	未标注；含 return	—

Models/OpenAI/IP_predictions/homopolymer_IP_4o_pos.py

函数/类	行号	参数	返回	静态调用（前 14 项）
construct_prompt	13–25	PSMILES, phyiscal_property, units_requested, use_common_name	未标注；含 return	—

Models/OpenAI/IP_predictions/homopolymer_IP_o3_neg.py

函数/类	行号	参数	返回	静态调用（前 14 项）
construct_prompt	13–25	PSMILES, phyiscal_property, units_requested, use_common_name	未标注；含 return	—

Models/OpenAI/IP_predictions/homopolymer_IP_o3_pos.py

函数/类	行号	参数	返回	静态调用（前 14 项）
construct_prompt	14–26	PSMILES, phyiscal_property, units_requested, use_common_name	未标注；含 return	—

Models/RDKit_RF/predict_rf_ensemble.py

函数/类	行号	参数	返回	静态调用（前 14 项）
calc_all_descriptors	15–19	psmi, missing	未标注；含 return	float, Chem.MolFromSmiles, CalcMolDescriptors
sanitize	22–37	X	pd.DataFrame	pd.DataFrame(X).apply, X.replace, X.drop, pd.DataFrame
convert_file	40–51	csv_path	未标注；含 return	pd.read_csv, df['smiles'].apply(calc_all_descriptors).apply, sanitize, ValueError, len, df[y_cols].to_numpy, df['smiles'].apply
align_columns	54–59	X, train_columns	pd.DataFrame	—
find_fold_models	63–73	root	未标注；含 return	root.resolve, sorted, root.name.startswith, (root / 'rf.joblib').exists, root.glob, root.rglob
load_train_columns	76–93	fold_dir	未标注；含 return	p.exists, load, hasattr, RuntimeError, list, open, json.load
main	96–178		未标注；无显式 return	argparse.ArgumentParser, ap.add_argument, ap.parse_args, find_fold_models, convert_file, load_train_columns, align_columns, np.stack, P.mean, P.std, df_raw.copy, out_df.to_csv, print, FileNotFoundError

Models/RDKit_RF/train_rf.py

函数/类	行号	参数	返回	静态调用（前 14 项）
sanitize_features	27–49	X	tuple[pd.DataFrame, list[str]]	X.drop, pd.DataFrame(X).apply, X.replace, X.columns[X.isna().all()].tolist, print, pd.DataFrame, X.isna().all, ', '.join, X.isna
load_and_prepare_data	52–95	data_path	未标注；含 return	pd.read_csv, sanitize_features, X.columns.tolist, len, print, sys.exit, df[col].nunique, target_cols.append, feature_cols.append, df.columns.tolist
train_random_forest	98–117	X_train, y_train, random_state	未标注；含 return	Pipeline, pipeline.fit, y_train.ravel, SimpleImputer, RandomForestRegressor
evaluate_model	120–132	model, X, y, split_name	未标注；含 return	model.predict, y.ravel, float, len, np.sqrt, mean_absolute_error, mean_squared_error
main	135–281		未标注；无显式 return	argparse.ArgumentParser, parser.add_argument, parser.parse_args, load_and_prepare_data, args.save_dir.mkdir, range, print, args.data_path.exists, sys.exit, open, json.dump, fold_dir.mkdir, train_test_split, train_random_forest

Models/RDKit_RF/train_rf_scaling.py

函数/类	行号	参数	返回	静态调用（前 14 项）
json_default	21–24	value	未标注；含 return	hasattr, TypeError, value.item, type
parse_args	27–34		argparse.Namespace	argparse.ArgumentParser, parser.add_argument, parser.parse_args
load_row_indices	37–43	path, expected_rows	np.ndarray	pd.read_csv, df['row_index'].to_numpy,

				ValueError, len
train_random_forest	46–62	x_train, y_train, random_state, n_estimators	Pipeline	Pipeline, model.fit, y_train.ravel, SimpleImputer, RandomForestRegressor
evaluate	65–78	model, x, y, split	tuple[dict, np.ndarray]	np.asarray(model.predict(x)).reshape, np.asarray(y).reshape, float, mean_squared_error, np.asarray, int, model.predict, np.sqrt, mean_absolute_error, len
write_predictions	81–89	path, original_indices, y_true, y_pred	None	path.parent.mkdir, pd.DataFrame({'row_index': original_indices.astype(np.int64), 'y_true': np.asarray(y_true).reshape(-1), 'y_pred': np.asarray(y_pred).reshape(-1)}).to_csv, pd.DataFrame, original_indices.astype, np.asarray(y_true).reshape, np.asarray(y_pred).reshape, np.asarray
align_columns	92–96	x, feature_cols	pd.DataFrame	—
train_one_split	99–148	split_dir, output_dir, seed, n_estimators	dict	load_and_prepare_data, align_columns, load_row_indices, train_random_forest, output_dir.mkdir, dump, len, evaluate, write_predictions, open, json.dump
train_size_from_dir	151–152	path	int	int, path.name.split
write_summary	155–179	rows, save_root	None	pd.DataFrame, detailed.to_csv, detailed.groupby(['model', 'train_size', 'split'], as_index=False).agg(rmse_mean=('rmse', 'mean'), rmse_std=('rmse', lambda x: float(np.std(x, ddof=0))), mae_mean=('mae', 'mean'), mae_std=('mae',

				lambda x: float(np.std(x, ddof=0))), n_folds=('fold', 'nunique')).sort_values, aggregate.to_csv, open, json.dump, detailed.groupby(['model', 'train_size', 'split'], as_index=False).agg, detailed.to_dict, aggregate.to_dict, detailed.groupby, float, np.std
main	182–222		None	parse_args, args.split_root.resolve, args.save_root.resolve, save_root.mkdir, sorted, write_summary, print, split_root.glob, int, ValueError, fold_dir.is_dir, fold_dir.glob, train_size_from_dir, train_one_split

Models/polyBERT/pBERT_predict_ensemble.py

函数/类	行号	参数	返回	静态调用（前 14 项）
FeedForwardNet	33–56	class	类实例	—
FeedForwardNet.__init__	34–53	self, input_dim, cfg	未标注；无显式 return	super().__init__, range, nn.Sequential, nn.Linear, super, nn.ReLU, nn.Dropout
FeedForwardNet.forward	55–56	self, x	未标注；含 return	self.net
XOnlyDataset	59–67	class	类实例	—
XOnlyDataset.__init__	60–61	self, X	未标注；无显式 return	torch.tensor
XOnlyDataset.__len__	63–64	self	未标注；含 return	—
XOnlyDataset.__getitem__	66–67	self, idx	未标注；含 return	—
predict_one_checkpoint	71–115	ckpt_path, X_raw, batch_size, device	np.ndarray	torch.no_grad, torch.load, FeedForwardNet(input_dim=input_dim, cfg=cfg).to, model.load_state_dict, model.eval, ckpt['feature_mean'].detach().cpu().numpy, ckpt['feature_std'].detach().cpu().nu

				mpy, XOnlyDataset, DataLoader, torch.cat(preds, dim=0).numpy, str(cfg.get('dataset_type', 'regression')).lower, xb.to, model, preds.reshape
load_csv_as_array	118–129	csv_path, x_cols, y_col, skip_header	Tuple[np.ndarray, Optional[np.ndarray]]	np.genfromtxt
main	132–206		未标注；无显式 return	argparse.ArgumentParser, ap.add_argument, ap.parse_args, Path, sorted, load_csv_as_array, np.arange, torch.load, out_path.parent.mkdir, print, Path().glob, FileNotFoundError, np.isfinite, str(cfg0.get('dataset_type', 'regression')).lower

Models/polyBERT/train_pBERT.py

函数/类	行号	参数	返回	静态调用（前 14 项）
make_out_dir_from_dataset	44–55	dataset_path, anchor	未标注；含 return	Path(dataset_path).resolve, parts.index, Path(*parts[i + 1:]).with_suffix, ValueError, Path
ArrayDataset	58–78	class	类实例	—
ArrayDataset.__init__	59–65	self, X, y	未标注；无显式 return	torch.tensor, torch.tensor(y, dtype=torch.float32).view
ArrayDataset.set_norm	67–69	self, mean, std	未标注；无显式 return	—
ArrayDataset.__len__	71–71	self	未标注；含 return	—
ArrayDataset.__getitem__	73–78	self, idx	未标注；含 return	—
FeedForwardNet	81–108	class	类实例	—
FeedForwardNet.__init__	82–105	self, input_dim, cfg	未标注；无显式 return	super().__init__, range, nn.Sequential, nn.Linear, super, nn.ReLU, nn.Dropout
FeedForwardNet.forward	107–108	self, x	未标注；含 return	self.net
evaluate	111–138	model, loader, criterion, device, is_classification	未标注；含 return	torch.no_grad, model.eval, math.sqrt, model, criterion, xb.size, (out.argmax(dim=

				1) == yb.view(-1).long()).sum().item, max, torch.sum((pred - yb) ** 2).item, xb.to, yb.to, yb.view(-1).long, loss.item, (out.argmax(dim=1) == yb.view(-1).long()).sum
predict_with_indices	141–155	model, loader, device, is_classification	未标注；含 return	torch.no_grad, model.eval, torch.cat(preds, dim=0).numpy, torch.cat(ys, dim=0).numpy, torch.cat(idxs, dim=0).numpy, xb.to, model, preds.append, ys.append, idxs.append, out.argmax(dim=1, keepdim=True).float, out.detach().cpu, yb.detach().cpu, idx.detach().cpu
train_one_epoch	158–171	model, loader, criterion, optimizer, device, scheduler	未标注；含 return	model.train, optimizer.zero_grad, model, criterion, loss.backward, optimizer.step, len, xb.to, yb.to, scheduler.step, loss.item, xb.size
run_repeated_experiments_from_arrays	173–367	X, y, cfg, out_dir, base_seed	未标注；含 return	ArrayDataset, len, int, Path, out_root.mkdir, range, print, torch.cuda.is_available, cfg['dataset_type'].lower, fold_dir.mkdir, torch.manual_seed, torch.Generator().manual_seed, random_split, torch.as_tensor
main	369–468		未标注；无显式 return	argparse.ArgumentParser, parser.add_argument, parser.parse_args, np.genfromtxt, np.ravel, np.isfinite, args.save_dir.mkdir, run_repeated_experiments_from_arrays, print,

				args.data_path.exists, sys.exit, str, len

Models/polyBERT/train_pBERT_scaling.py

函数/类	行号	参数	返回	静态调用（前 14 项）
json_default	22–25	value	未标注；含 return	hasattr, TypeError, value.item, type
parse_args	28–37		argparse.Namespace	argparse.ArgumentParser, parser.add_argument, parser.parse_args
load_polybert_csv	40–49	path	tuple[np.ndarray, np.ndarray]	np.genfromtxt, np.atleast_2d, ValueError, np.isfinite(y).all, np.isfinite
load_row_indices	52–58	path, expected_rows	np.ndarray	pd.read_csv, df['row_index'].to_numpy, ValueError, len
metrics_from_arrays	61–71	y_true, y_pred, split	dict[str, float | int | str]	np.asarray(y_true, dtype=float).reshape, np.asarray(y_pred, dtype=float).reshape, float, np.mean, int, np.asarray, math.sqrt, len, np.abs
predict_split	74–77	model, dataset, batch_size, device	tuple[np.ndarray, np.ndarray, np.ndarray]	DataLoader, predict_with_indices, pred.reshape, y.reshape, local_idx.reshape
write_predictions	80–88	path, original_indices, y_true, y_pred	None	path.parent.mkdir, pd.DataFrame({'row_index': original_indices.astype(np.int64), 'y_true': y_true.reshape(-1), 'y_pred': y_pred.reshape(-1)}).to_csv, pd.DataFrame, original_indices.astype, y_true.reshape, y_pred.reshape
train_one_split	91–206	split_dir, output_dir, cfg, seed, device, quiet	dict	load_polybert_csv, load_row_indices, torch.manual_seed, ArrayDataset, train_ds.X.mean, train_ds.X.std, torch.Generator().manual_seed, DataLoader, FeedForwardNet(input_dim=train_x.shape[1], cfg=cfg).to,

				nn.MSELoss, torch.optim.Adam, max, min, torch.optim.lr_scheduler.OneCycleLR
train_size_from_dir	209–210	path	int	int, path.name.split
write_summary	213–239	rows, save_root	None	pd.DataFrame, detailed.to_csv, detailed.groupby(['model', 'train_size', 'split'], as_index=False).agg(rmse_mean=('rmse', 'mean'), rmse_std=('rmse', lambda x: float(np.std(x, ddof=0))), mae_mean=('mae', 'mean'), mae_std=('mae', lambda x: float(np.std(x, ddof=0))), n_folds=('fold', 'nunique')).sort_values, aggregate.to_csv, open, json.dump, detailed.groupby(['model', 'train_size', 'split'], as_index=False).agg, detailed.to_dict, aggregate.to_dict, detailed.groupby, float, np.std
main	242–291		None	parse_args, args.split_root.resolve, args.save_root.resolve, save_root.mkdir, dict, sorted, write_summary, print, split_root.glob, int, ValueError, torch.cuda.is_available, fold_dir.is_dir, fold_dir.glob

Models/polymer_periodic_graph/Polymer_Model/chemprop/__init__.py

无可解析顶层函数/类，可能为脚本式逻辑或纯导入文件。

Models/polymer_periodic_graph/Polymer_Model/chemprop/_version.py

无可解析顶层函数/类，可能为脚本式逻辑或纯导入文件。

Models/polymer_periodic_graph/Polymer_Model/chemprop/args.py

函数/类	行号	参数	返回	静态调用（前 14 项）
get_checkpoint_paths	19–59	checkpoint_path, checkpoint_paths, checkpoint_dir, ext	Optional[List[str]]	sum, ValueError, os.walk, len, fname.endswith, checkpoint_paths.append, os.path.join
CommonArgs	62–214	class	类实例	—
CommonArgs.__init__	114–118	self	未标注；无显式 return	super(CommonArgs, self).__init__, super
CommonArgs.device	121–126	self	torch.device	torch.device
CommonArgs.device	129–131	self, device	None	—
CommonArgs.cuda	134–136	self	bool	torch.cuda.is_available
CommonArgs.cuda	139–140	self, cuda	None	—
CommonArgs.features_scaling	143–148	self	bool	—
CommonArgs.features_scaling	151–152	self, features_scaling	None	—
CommonArgs.atom_features_size	155–157	self	int	—
CommonArgs.atom_features_size	160–161	self, atom_features_size	None	—
CommonArgs.atom_descriptors_size	164–166	self	int	—
CommonArgs.atom_descriptors_size	169–170	self, atom_descriptors_size	None	—
CommonArgs.bond_features_size	173–175	self	int	—
CommonArgs.bond_features_size	178–179	self, bond_features_size	None	—
CommonArgs.configure	181–183	self	None	self.add_argument, list, get_available_features_generators, range, torch.cuda.device_count
CommonArgs.process_args	185–214	self	None	get_checkpoint_paths, set_cache_mol, ValueError, NotImplementedError, empty_cache
TrainArgs	217–628	class	类实例	—
TrainArgs.__init__	421–428	self	None	super(TrainArgs, self).__init__, super
TrainArgs.metrics	431–433	self	List[str]	—
TrainArgs.minimize_score	436–438	self	bool	—
TrainArgs.use_input_features	441–443	self	bool	—
TrainArgs.num_lr	446–448	self	int	—

s				
TrainArgs.crossval_index_sets	451–453	self	List[List[List[int]]]	—
TrainArgs.task_names	456–458	self	List[str]	—
TrainArgs.task_names	461–462	self, task_names	None	—
TrainArgs.num_tasks	465–467	self	int	len
TrainArgs.features_size	470–472	self	int	—
TrainArgs.features_size	475–476	self, features_size	None	—
TrainArgs.train_data_size	479–481	self	int	—
TrainArgs.train_data_size	484–485	self, train_data_size	None	—
TrainArgs.atom_descriptor_scaling	488–493	self	bool	—
TrainArgs.bond_feature_scaling	496–501	self	bool	—
TrainArgs.process_args	503–628	self	None	super(TrainArgs, self).process_args, chemprop.data.utils.preprocess_smiles_columns, TemporaryDirectory, len, ValueError, NotImplementedError, super, open, json.load, config.items, any, pickle.load, sum, min
PredictArgs	631–661	class	类实例	—
PredictArgs.ensemble_size	646–648	self	int	len
PredictArgs.process_args	650–661	self	None	super(PredictArgs, self).process_args, chemprop.data.utils.preprocess_smiles_columns, ValueError, super, len
InterpretArgs	664–701	class	类实例	—
InterpretArgs.process_args	684–701	self	None	super(InterpretArgs, self).process_args, chemprop.data.utils.preprocess_smiles_columns, ValueError, super, len
FingerprintArgs	704–708	class	类实例	—
HyperoptArgs	711–737	class	类实例	—
HyperoptArgs.process_args	730–737	self	None	super(HyperoptArgs, self).process_args, super
SklearnTrainArgs	740–756	class	类实例	—
SklearnPredictAr	759–793	class	类实例	—

gs				
SklearnPredictArgs.process_args	779–793	self	None	chemprop.data.utils.preprocess_smiles_columns, get_checkpoint_paths

Models/polymer_periodic_graph/Polymer_Model/chemprop/constants.py

无可解析顶层函数/类，可能为脚本式逻辑或纯导入文件。

Models/polymer_periodic_graph/Polymer_Model/chemprop/data/__init__.py

无可解析顶层函数/类，可能为脚本式逻辑或纯导入文件。

Models/polymer_periodic_graph/Polymer_Model/chemprop/data/data.py

函数/类	行号	参数	返回	静态调用（前 14 项）
cache_graph	21–23		bool	—
set_cache_graph	26–29	cache_graph	None	—
empty_cache	32–35		未标注；无显式 return	SMILES_TO_GRAPH.clear, SMILES_TO_MOL.clear
cache_mol	43–45		bool	—
set_cache_mol	48–51	cache_mol	None	—
MoleculeDatapoint	54–228	class	类实例	—
MoleculeDatapoint.__init__	57–146	self, smiles, targets, row, data_weight, features, features_generator, phase_features, atom_features, atom_descriptors, bond_featu	未标注；无显式 return	is_reaction, is_explicit_h, ValueError, np.array, np.where, get_features_generator, np.isnan, self.features.extend, m.GetNumHeavyAtoms, features_generator, m[0].GetNumHeavyAtoms, np.zeros, len, Chem.MolFromSmiles
MoleculeDatapoint.mol	149–157	self	Union[List[Chem.Mol], List[Tuple[Chem.Mol, Chem.Mol]]]	make_mols, cache_mol, zip
MoleculeDatapoint.number_of_molecules	160–166	self	int	len
MoleculeDatapoint.set_features	168–174	self, features	None	—
MoleculeDatapoint.set_atom_descriptors	176–182	self, atom_descriptors	None	—
MoleculeDatapoint.set_atom_features	184–190	self, atom_features	None	—
MoleculeDatapoint.set_bond_features	192–198	self, bond_features	None	—
MoleculeDatapoi	200–206	self, features	None	np.append

nt.extend_features				
MoleculeDatapoint.num_tasks	208–214	self	int	len
MoleculeDatapoint.set_targets	216–222	self, targets	未标注；无显式 return	—
MoleculeDatapoint.reset_features_and_targets	224–228	self	None	—
MoleculeDataset	231–532	class	类实例	—
MoleculeDataset.__init__	234–240	self, data	未标注；无显式 return	Random
MoleculeDataset.smiles	242–252	self, flatten	Union[List[str], List[List[str]]]	—
MoleculeDataset.mols	254–264	self, flatten	Union[List[Chem.Mol], List[List[Chem.Mol]], List[Tuple[Chem.Mol,	—
MoleculeDataset.number_of_molecules	267–273	self	int	len
MoleculeDataset.batch_graph	275–312	self	List[BatchMolGraph]	zip, mol_graphs.append, BatchMolGraph, mol_graphs_list.append, range, MolGraph, cache_graph, len, NotImplementedError
MoleculeDataset.features	314–323	self	List[np.ndarray]	len
MoleculeDataset.phase_features	325–334	self	List[np.ndarray]	len
MoleculeDataset.atom_features	336–346	self	List[np.ndarray]	len
MoleculeDataset.atom_descriptors	348–358	self	List[np.ndarray]	len
MoleculeDataset.bond_features	360–370	self	List[np.ndarray]	len
MoleculeDataset.data_weights	372–376	self	List[float]	—
MoleculeDataset.targets	378–384	self	List[List[Optional[float]]]	—
MoleculeDataset.num_tasks	386–392	self	int	self._data[0].num_tasks, len
MoleculeDataset.features_size	394–400	self	int	len
MoleculeDataset.atom_descriptors_size	402–409	self	int	len
MoleculeDataset.atom_features_size	411–418	self	int	len
MoleculeDataset.bond_features_size	420–427	self	int	len
MoleculeDataset.normalize_features	429–480	self, scaler, replace_nan_token, scale_atom_descriptors, scale_bond_features	StandardScaler	StandardScaler, scaler.fit, len, np.vstack, d.set_atom_descriptors, scaler.transform, d.set_atom_featur

				es, d.set_bond_features, d.set_features, d.raw_features.reshape
MoleculeDataset.normalize_targets	482–498	self	StandardScaler	StandardScaler().fit, scaler.transform(targets).tolist, self.set_targets, StandardScaler, scaler.transform
MoleculeDataset.set_targets	500–509	self, targets	None	range, len, self._data[i].set_targets
MoleculeDataset.reset_features_and_targets	511–514	self	None	d.reset_features_and_targets
MoleculeDataset.__len__	516–522	self	int	len
MoleculeDataset.__getitem__	524–532	self, item	Union[MoleculeDatapoint, List[MoleculeDatapoint]]	—
MoleculeSampler	535–589	class	类实例	—
MoleculeSampler.__init__	538–569	self, dataset, class_balance, shuffle, seed	未标注；无显式 return	super(Sampler, self).__init__, Random, np.arange, np.array, indices[has_active].tolist, indices[~has_active].tolist, len, super, min, any
MoleculeSampler.__iter__	571–585	self	Iterator[int]	iter, list, self._random.shuffle, range, zip, len
MoleculeSampler.__len__	587–589	self	int	—
construct_molecule_batch	592–605	data	MoleculeDataset	MoleculeDataset, data.batch_graph
MoleculeDataLoader	608–678	class	类实例	—
MoleculeDataLoader.__init__	611–657	self, dataset, batch_size, num_workers, class_balance, shuffle, seed	未标注；无显式 return	MoleculeSampler, super(MoleculeDataLoader, self).__init__, threading.current_thread, threading.main_thread, super
MoleculeDataLoader.targets	660–669	self	List[List[Optional[float]]]	ValueError
MoleculeDataLoader.iter_size	672–674	self	int	len
MoleculeDataLoader.__iter__	676–678	self	Iterator[MoleculeDataset]	super(MoleculeDataLoader, self).__iter__, super
make_mols	681–694	smiles, reaction, keep_h	未标注；含 return	make_mol, s.split

Models/polymer_periodic_graph/Polymer_Model/chemprop/data/scaffold.py

函数/类	行号	参数	返回	静态调用（前 14

				项）
generate_scaffold	15–29	mol, include_chirality	str	isinstance, MurckoScaffold.MurckoScaffoldSmiles, make_mol
scaffold_to_smiles	32–50	mols, use_indices	Dict[str, Union[Set[str], Set[int]]]	defaultdict, tqdm, enumerate, generate_scaffold, len, scaffolds[scaffold].add
scaffold_split	53–130	data, sizes, balanced, seed, logger	Tuple[MoleculeDataset, MoleculeDataset, MoleculeDataset]	scaffold_to_smiles, Random, sum, ValueError, data.mols, list, random.seed, random.shuffle, sorted, logger.debug, log_scaffold_stats, MoleculeDataset, len, scaffold_to_indices.values
log_scaffold_stats	133–173	data, index_sets, num_scaffolds, num_labels, logger	List[Tuple[List[float], List[int]]]	sorted, enumerate, logger.debug, np.array, stats.append, warnings.catch_warnings, warnings.simplefilter, np.count_nonzero, len, np.nanmean, zip, np.isnan

Models/polymer_periodic_graph/Polymer_Model/chemprop/data/scaler.py

函数/类	行号	参数	返回	静态调用（前 14 项）
StandardScaler	6–63	class	类实例	—
StandardScaler.__init__	13–21	self, means, stds, replace_nan_token	未标注；无显式 return	—
StandardScaler.fit	23–37	self, X	'StandardScaler'	np.array(X).astype, np.nanmean, np.nanstd, np.where, np.isnan, np.zeros, np.ones, np.array
StandardScaler.transform	39–50	self, X	np.ndarray	np.array(X).astype, np.where, np.isnan, np.array
StandardScaler.inverse_transform	52–63	self, X	np.ndarray	np.array(X).astype, np.where, np.isnan, np.array

Models/polymer_periodic_graph/Polymer_Model/chemprop/data/utils.py

函数/类	行号	参数	返回	静态调用（前 14 项）
preprocess_smiles_columns	19–50	path, smiles_columns, number_of_molec	List[Optional[str]]	os.path.isfile, get_header, isinstance, any,

		ules		len, ValueError
get_task_names	53–85	path, smiles_columns, target_columns, ignore_columns	List[str]	get_header, set, isinstance, preprocess_smiles_columns
get_header	88–98	path	List[str]	open, next, csv.reader
get_data_weights	101–119	path	List[float]	open, csv.reader, next, sum, len, min, ValueError, weights.append, float
get_smiles	122–155	path, smiles_columns, header, flatten	Union[List[str], List[List[str]]]	ValueError, isinstance, preprocess_smiles_columns, open, csv.DictReader, csv.reader
filter_invalid_smiles	158–168	data	MoleculeDataset	MoleculeDataset, tqdm, all, m.GetNumHeavyAtoms, isinstance, m[0].GetNumHeavyAtoms, m[1].GetNumHeavyAtoms
get_data	171–349	path, smiles_columns, target_columns, ignore_columns, skip_invalid_smiles, args, data_weights_path, features_path, features_genera	MoleculeDataset	isinstance, preprocess_smiles_columns, float, np.concatenate, load_features, get_data_weights, open, csv.DictReader, enumerate, MoleculeDataset, len, filter_invalid_smiles, features_data.append, np.array
get_data_from_smiles	352–383	smiles, skip_invalid_smiles, logger, features_generator	MoleculeDataset	MoleculeDataset, len, filter_invalid_smiles, MoleculeDatapoint, debug, OrderedDict
split_data	386–543	data, split_type, sizes, seed, num_folds, args, logger	Tuple[MoleculeDataset, MoleculeDataset, MoleculeDataset]	Random, ValueError, range, tuple, data_split.append, MoleculeDataset, random.shuffle, zip, len, sum, np.repeat, open, split_indices.extend, np.arange
get_class_sizes	546–574	data	List[List[float]]	data.targets, range, len, class_sizes.append, set, ValueError, data.num_tasks, valid_targets[task_num].append, np.unique,

				np.count_nonzero, float, print
validate_dataset_type	578–593	data, dataset_type	None	ValueError, data.targets
validate_data	596–653	data_path	Set[str]	set, get_header, Chem.MolFromSmiles, tqdm, open, csv.reader, next, len, errors.add, num_tasks_set.pop, np.unique, smiles.append, targets.append, float

Models/polymer_periodic_graph/Polymer_Model/chemprop/features/__init__.py

无可解析顶层函数/类，可能为脚本式逻辑或纯导入文件。

Models/polymer_periodic_graph/Polymer_Model/chemprop/features/features_generators.py

函数/类	行号	参数	返回	静态调用（前 14 项）
register_features_generator	15–26	features_generator_name	Callable[[FeaturesGenerator], FeaturesGenerator]	—
get_features_generator	29–40	features_generator_name	FeaturesGenerator	ValueError
get_available_features_generators	43–45		List[str]	list, FEATURES_GENERATOR_REGISTRY.keys
morgan_binary_features_generator	53–69	mol, radius, num_bits	np.ndarray	register_features_generator, AllChem.GetMorganFingerprintAsBitVect, np.zeros, DataStructs.ConvertToNumpyArray, Chem.MolFromSmiles, type
morgan_counts_features_generator	73–89	mol, radius, num_bits	np.ndarray	register_features_generator, AllChem.GetHashedMorganFingerprint, np.zeros, DataStructs.ConvertToNumpyArray, Chem.MolFromSmiles, type

Models/polymer_periodic_graph/Polymer_Model/chemprop/features/featurization.py

函数/类	行号	参数	返回	静态调用（前 14 项）
Featurization_parameters	15–54	class	类实例	—
Featurization_parameters.__init__	19–54	self	None	list, range, sum, len, self.ATOM_FEATURES.values
reset_featurizatio	60–70	logger	None	debug,

n_parameters				Featurization_parameters
get_atom_fdim	73–80	overwrite_default_atom	int	—
set_explicit_h	83–89	explicit_h	None	—
set_reaction	92–104	reaction, mode	None	—
is_explicit_h	107–109		bool	—
is_reaction	112–114		bool	—
reaction_mode	117–119		str	—
set_extra_atom_fdim	122–124	extra	未标注；无显式 return	—
get_bond_fdim	127–142	atom_messages, overwrite_default_bond, overwrite_default_atom	int	get_atom_fdim
set_extra_bond_fdim	145–147	extra	未标注；无显式 return	—
onek_encoding_unk	150–163	value, choices	List[int]	choices.index, len
cgcnn_atom_features	166–187	atom, functional_groups	List[Union[bool, int, float]]	print, atom.GetSymbol, onek_encoding_unk, np.min, list, Element, range, np.where, np.linspace
atom_features	189–210	atom, functional_groups	List[Union[bool, int, float]]	atom.GetAtomicNum, onek_encoding_unk, atom.GetMass, int, atom.GetIsAromatic, atom.GetHybridization, atom.GetTotalNumHs, atom.GetFormalCharge, atom.GetChiralTag, atom.GetTotalDegree
atom_features_zeros	213–225	atom	List[Union[bool, int, float]]	onek_encoding_unk, atom.GetAtomicNum
bond_features	228–249	bond	List[Union[bool, int, float]]	bond.GetBondType, onek_encoding_unk, int, list, bond.GetIsConjugated, bond.IsInRing, bond.GetStereo, range
bond_features_new	251–275	bond, bond_length	List[Union[bool, int, float]]	onek_encoding_unk, bond.GetBondType, np.min, list, int, range, bond.GetIsConjugated, bond.IsInRing, bond.GetStereo,

				np.where, np.linspace
cgcnn_bond_features	277–289	bond	List[Union[bool, int, float]]	print, onek_encoding_unk, np.min, list, range, np.where, np.linspace
map_reac_to_prod	291–321	mol_reac, mol_prod	未标注；含 return	set, mol_prod.GetAtoms, mol_reac.GetAtoms, atom.GetAtomMapNum, atom.GetIdx, only_prod_ids.append, only_reac_ids.append
MolGraph	324–539	class	类实例	—
MolGraph.__init__	341–539	self, mol, atom_features_extra, bond_features_extra, overwrite_default_atom_features, overwrite_default_bond_features	未标注；无显式 return	is_reaction, is_explicit_h, reaction_mode, Chem.AddHs, Chem.MolToMolBlock, output_mol_array.split, mol.GetNumAtoms, mol.GetAtoms, np.array, type, atom.GetNeighbors, len, range, map_reac_to_prod
BatchMolGraph	541–662	class	类实例	—
BatchMolGraph.__init__	556–603	self, mol_graphs	未标注；无显式 return	get_atom_fdim, get_bond_fdim, max, torch.FloatTensor, torch.LongTensor, f_atoms.extend, f_bonds.extend, range, self.a_scope.append, self.b_scope.append, a2b.append, b2a.append, b2revb.append, len
BatchMolGraph.get_components	605–633	self, atom_messages	Tuple[torch.FloatTensor, torch.FloatTensor, torch.LongTensor, tor	get_bond_fdim
BatchMolGraph.get_b2b	635–647	self	torch.LongTensor	(b2b != self.b2revb.unsqueeze(1).repeat(1, b2b.size(1))).long, self.b2revb.unsqueeze(1).repeat, b2b.size, self.b2revb.unsqueeze
BatchMolGraph.get_a2a	649–662	self	torch.LongTensor	—
mol2graph	665–685	mols,	BatchMolGraph	BatchMolGraph,

		atom_features_batch, bond_features_batch, overwrite_default_atom_features, overwrite_default_bond_features		MolGraph, zip_longest

Models/polymer_periodic_graph/Polymer_Model/chemprop/features/featurization_copy.py

函数/类	行号	参数	返回	静态调用（前 14 项）
Featurization_parameters	11–47	class	类实例	—
Featurization_parameters.__init__	15–47	self	None	list, range, sum, len, self.ATOM_FEATURES.values
reset_featurization_parameters	53–63	logger	None	debug, Featurization_parameters
get_atom_fdim	66–73	overwrite_default_atom	int	—
set_explicit_h	76–82	explicit_h	None	—
set_reaction	85–97	reaction, mode	None	—
is_explicit_h	100–102		bool	—
is_reaction	105–107		bool	—
reaction_mode	110–112		str	—
set_extra_atom_fdim	115–117	extra	未标注；无显式 return	—
get_bond_fdim	120–135	atom_messages, overwrite_default_bond, overwrite_default_atom	int	get_atom_fdim
set_extra_bond_fdim	138–140	extra	未标注；无显式 return	—
onek_encoding_unk	143–156	value, choices	List[int]	choices.index, len
atom_features	159–180	atom, functional_groups	List[Union[bool, int, float]]	onek_encoding_unk, atom.GetMass, int, atom.GetIsAromatic, atom.GetHybridization, atom.GetTotalNumHs, atom.GetFormalCharge, atom.GetChiralTag, atom.GetTotalDegree, atom.GetAtomicNum
atom_features_zeros	183–195	atom	List[Union[bool, int, float]]	onek_encoding_unk, atom.GetAtomicNum
bond_features	198–219	bond	List[Union[bool, int, float]]	bond.GetBondType, onek_encoding_u

				nk, int, list, bond.GetIsConjugated, bond.IsInRing, bond.GetStereo, range
map_reac_to_prod	222–252	mol_reac, mol_prod	未标注；含 return	set, mol_prod.GetAtoms, mol_reac.GetAtoms, atom.GetAtomMapNum, atom.GetIdx, only_prod_ids.append, only_reac_ids.append
MolGraph	255–453	class	类实例	—
MolGraph.__init__	272–453	self, mol, atom_features_extra, bond_features_extra, overwrite_default_atom_features, overwrite_default_bond_features	未标注；无显式 return	is_reaction, is_explicit_h, reaction_mode, type, len, range, map_reac_to_prod, mol_reac.GetNumAtoms, make_mol, atom_features, ValueError, self.a2b.append, NotImplementedError, mol.GetAtoms
BatchMolGraph	455–576	class	类实例	—
BatchMolGraph.__init__	470–517	self, mol_graphs	未标注；无显式 return	get_atom_fdim, get_bond_fdim, max, torch.FloatTensor, torch.LongTensor, f_atoms.extend, f_bonds.extend, range, self.a_scope.append, self.b_scope.append, a2b.append, b2a.append, b2revb.append, len
BatchMolGraph.get_components	519–547	self, atom_messages	Tuple[torch.FloatTensor, torch.FloatTensor, torch.LongTensor, tor	get_bond_fdim
BatchMolGraph.get_b2b	549–561	self	torch.LongTensor	(b2b != self.b2revb.unsqueeze(1).repeat(1, b2b.size(1))).long, self.b2revb.unsqueeze(1).repeat, b2b.size, self.b2revb.unsqueeze
BatchMolGraph.get_a2a	563–576	self	torch.LongTensor	—
mol2graph	579–599	mols, atom_features_batch,	BatchMolGraph	BatchMolGraph, MolGraph, zip_longest

		bond_features_batch, overwrite_default_atom_features, overwrite_default_bond_features		

Models/polymer_periodic_graph/Polymer_Model/chemprop/features/utils.py

函数/类	行号	参数	返回	静态调用（前 14 项）
save_features	11–18	path, features	None	np.savez_compressed
load_features	21–57	path	np.ndarray	os.path.splitext, np.load, open, csv.reader, next, np.array, ValueError, float, np.squeeze, pickle.load, feat.todense
load_valid_atom_or_bond_features	60–107	path, smiles	List[np.ndarray]	os.path.splitext, np.load, pd.read_pickle, features_df.apply(lambda x: np.stack(x.tolist(), axis=1), axis=1).tolist, PandasTools.LoadSDF(path).drop(['ID', 'ROMol'], axis=1).set_index, features_df.reindex, features_df.isnull().any().any, features_df.applymap, ValueError, features_df.apply(lambda x: np.concatenate(x.tolist(), axis=1), axis=1).tolist, features_df.apply, PandasTools.LoadSDF(path).drop, features_df.index.duplicated, features_df.iloc[0, :].apply(lambda x: isinstance(x, str) and ',' in x).to_list

Models/polymer_periodic_graph/Polymer_Model/chemprop/hyperopt_utils.py

函数/类	行号	参数	返回	静态调用（前 14 项）
merge_trials	13–39	trials, new_trials_data	Trials	len, max, Trials().new_trial_docs, hyperopt_trial[0]['misc']['idxs'].keys, trials.insert_trial_docs, trials.refresh, Trials

load_trials	42–66	dir_path, previous_trials	Trials	Trials, os.path.join, merge_trials, os.listdir, open, pickle.load
save_trials	69–80	dir_path, trials, hyperopt_seed	None	os.listdir, pickle.dump, ValueError, open, os.path.join
get_hyperopt_seed	83–113	seed, dir_path	int	os.path.join, os.path.exists, seeds.append, makedirs, int, ''.join, open, f.write, next, seeds.extend, map, seed_line.split
load_manual_trials	116–205	manual_trials_dirs, param_keys, hyperopt_args	Trials	enumerate, Trials, merge_trials, float, manual_trials_data.append, open, csv.reader, next, json.load, os.path.join, ValueError, getattr

Models/polymer_periodic_graph/Polymer_Model/chemprop/hyperparameter_optimization.py

函数/类	行号	参数	返回	静态调用（前 14 项）
hyperopt	31–156	args	None	timeit, create_logger, makedirs, range, load_trials, min, logger.info, load_manual_trials, deepcopy, hyperparams.items, cross_validate, MoleculeModel, param_count, np.isnan
chemprop_hyperopt	159–164		None	hyperopt, HyperoptArgs().parse_args, HyperoptArgs

Models/polymer_periodic_graph/Polymer_Model/chemprop/interpret.py

函数/类	行号	参数	返回	静态调用（前 14 项）
ChempropModel	17–75	class	类实例	—
ChempropModel.__init__	20–38	self, args	None	load_args, load_scalers, ValueError, NotImplementedError, load_checkpoint
ChempropModel.__call__	40–75	self, smiles, batch_size	List[List[float]]	get_data_from_smiles, MoleculeDataset, MoleculeDataLoader, sum, test_data.normaliz

				e_features, predict, sum_preds.append, len, range, np.array
MCTSNode	78–100	class	类实例	—
MCTSNode.__init__	81–94	self, smiles, atoms, W, N, P	None	set
MCTSNode.Q	96–97	self	float	—
MCTSNode.U	99–100	self, n	float	math.sqrt
find_clusters	103–130	mol	Tuple[List[Tuple[int, ...]], List[List[int]]]	mol.GetNumAtoms, mol.GetBonds, clusters.extend, range, bond.GetBeginAtom().GetIdx, bond.GetEndAtom().GetIdx, tuple, len, bond.IsInRing, clusters.append, Chem.GetSymmSSSR, atom_cls[atom].append, bond.GetBeginAtom, bond.GetEndAtom
__extract_subgraph	133–166	mol, selected_atoms	Tuple[Chem.Mol, List[int]]	set, Chem.RWMol, sorted, mol.GetAtomWithIdx, new_mol.GetAtomWithIdx, atom.SetAtomMapNum, atom.GetIdx, new_mol.RemoveAtom, new_mol.GetMol, len, roots.append, atom.SetIsAromatic, new_mol.GetAtoms, atom.GetNeighbors
extract_subgraph	169–200	smiles, selected_atoms	Tuple[str, List[int]]	Chem.MolFromSmiles, Chem.Kekulize, __extract_subgraph, Chem.MolToSmiles, mol.HasSubstructMatch
mcts_rollout	203–255	node, state_map, orig_smiles, clusters, atom_cls, nei_cls, scoring_function	float	sum, max, mcts_rollout, len, set, scoring_function, zip, extract_subgraph, enumerate, MCTSNode, node.children.append, x.Q, x.U
mcts	258–294	smiles, scoring_function,	List[MCTSNode]	Chem.MolFromSmiles, find_clusters,

		n_rollout, max_atoms, prop_delta		enumerate, range, MCTSNode, mol.GetNumAtoms, len, set, mcts_rollout, list, state_map.items
interpret	297–342	args	None	timeit, ChempropModel, get_smiles, get_header, print, ValueError, chemprop_model, len, scoring_function, mcts, min, sorted
chemprop_interpret	345–350		None	interpret, InterpretArgs().parse_args, InterpretArgs

Models/polymer_periodic_graph/Polymer_Model/chemprop/models/__init__.py

无可解析顶层函数/类，可能为脚本式逻辑或纯导入文件。

Models/polymer_periodic_graph/Polymer_Model/chemprop/models/model.py

函数/类	行号	参数	返回	静态调用（前 14 项）
MoleculeModel	14–185	class	类实例	—
MoleculeModel.__init__	17–39	self, args	未标注；无显式 return	super(MoleculeModel, self).__init__, self.create_encoder, self.create_ffn, initialize_weights, nn.Sigmoid, nn.Softmax, super
MoleculeModel.create_encoder	41–55	self, args	None	MPN, list(self.encoder.encoder.children())[0].parameters, self.encoder.parameters, list, self.encoder.encoder.children
MoleculeModel.create_ffn	57–121	self, args	None	nn.Dropout, get_activation_function, nn.Sequential, range, ffn.extend, ffn.append, nn.Linear, nn.Softplus, nn_exp, list, super(nn_exp, self).__init__, torch.exp, self.ffn.parameters, super
MoleculeModel.fingerprint	123–150	self, batch, features_batch, atom_descriptors_batch, atom_features_batch, bond_features_batch,	torch.FloatTensor	self.encoder, self.ffn[:-1], ValueError

		fingerprint_type		
MoleculeModel.forward	152–185	self, batch, features_batch, atom_descriptors_batch, atom_features_batch, bond_features_batch	torch.FloatTensor	self.ffn, self.encoder, self.sigmoid, output.reshape, self.multiclass_softmax, output.size

Models/polymer_periodic_graph/Polymer_Model/chemprop/models/mpn.py

函数/类	行号	参数	返回	静态调用（前 14 项）
MPNEncoder	14–150	class	类实例	—
MPNEncoder.__init__	17–64	self, args, atom_fdim, bond_fdim	未标注；无显式 return	super(MPNEncoder, self).__init__, nn.Dropout, get_activation_function, nn.Parameter, nn.Linear, torch.zeros, super
MPNEncoder.forward	66–150	self, mol_graph, atom_descriptors_batch	torch.FloatTensor	mol_graph.get_components, self.act_func, range, index_select_ND, nei_a_message.sum, torch.cat, self.dropout_layer, enumerate, torch.stack, torch.from_numpy(np.concatenate(atom_descriptors_batch, axis=0)).float().to, f_atoms.to, f_bonds.to, a2b.to, b2a.to
MPN	153–266	class	类实例	—
MPN.__init__	156–185	self, args, atom_fdim, bond_fdim	未标注；含 return	super(MPN, self).__init__, get_atom_fdim, get_bond_fdim, nn.ModuleList, super, MPNEncoder, range
MPN.forward	187–266	self, batch, features_batch, atom_descriptors_batch, atom_features_batch, bond_features_batch	torch.FloatTensor	reduce, type, torch.from_numpy(np.stack(features_batch)).float().to, torch.cat, len, NotImplementedError, enc, features_batch.view, range, mol2graph, torch.from_numpy(np.stack(features_batch)).float, zip, torch.from_numpy, np.stack

Models/polymer_periodic_graph/Polymer_Model/chemprop/nn_utils.py

函数/类	行号	参数	返回	静态调用（前 14 项）
compute_pnorm	11–18	model	float	math.sqrt, sum, p.norm().item, model.parameters, p.norm
compute_gnorm	21–28	model	float	math.sqrt, sum, p.grad.norm().item, model.parameters, p.grad.norm
param_count	31–38	model	int	sum, param.numel, model.parameters
param_count_all	40–47	model	int	sum, param.numel, model.parameters
index_select_ND	50–67	source, index	torch.Tensor	index.size, source.index_select, target.view, source.size, index.view
get_activation_function	70–99	activation	nn.Module	nn.ReLU, nn.LeakyReLU, nn.PReLU, nn.Tanh, nn.SELU, nn.ELU, ValueError
initialize_weights	102–112	model	None	model.parameters, param.dim, nn.init.constant_, nn.init.xavier_normal_
NoamLR	115–194	class	类实例	—
NoamLR.__init__	126–164	self, optimizer, warmup_epochs, total_epochs, steps_per_epoch, init_lr, max_lr, final_lr	未标注；无显式 return	len, np.array, (self.warmup_epochs * self.steps_per_epoch).astype, super(NoamLR, self).__init__, super
NoamLR.get_lr	166–172	self	List[float]	list
NoamLR.step	174–194	self, current_step	未标注；无显式 return	range

Models/polymer_periodic_graph/Polymer_Model/chemprop/rdkit.py

函数/类	行号	参数	返回	静态调用（前 14 项）
make_mol	3–16	s, keep_h	未标注；含 return	Chem.MolFromSmiles, Chem.SanitizeMol

Models/polymer_periodic_graph/Polymer_Model/chemprop/sklearn_predict.py

函数/类	行号	参数	返回	静态调用（前 14 项）
predict_sklearn	15–74	args	None	timeit, print, get_data, get_features_generator, tqdm, np.zeros,

				avg_preds.tolist, makedirs, zip, open, pickle.load, SklearnTrainArgs().from_dict, predict, np.array
sklearn_predict	77–82		None	predict_sklearn, SklearnPredictArgs().parse_args, SklearnPredictArgs

Models/polymer_periodic_graph/Polymer_Model/chemprop/sklearn_train.py

函数/类	行号	参数	返回	静态调用（前 14 项）
predict	20–57	model, model_type, dataset_type, features	List[List[float]]	model.predict, len, ValueError, model.predict_proba, type, model.decision_function, range
impute_sklearn	59–127	model, train_data, args, logger, threshold	List[float]	train_data.num_tasks, deepcopy, debug, trange, train_data.targets, len, zip, range, model.fit, predict, train_data.features, int, np.median, np.mean
single_task_sklearn	130–184	model, train_data, test_data, metrics, args, logger	List[float]	train_data.num_tasks, trange, zip, model.fit, predict, evaluate_predictions, scores[metric].append, train_data.features, train_data.targets, test_data.features, test_data.targets
multi_task_sklearn	187–247	model, train_data, test_data, metrics, args, logger	Dict[str, List[float]]	train_data.num_tasks, train_data.targets, model.fit, predict, evaluate_predictions, impute_sklearn, any, train_data.features, open, pickle.dump, ValueError, os.path.join, test_data.features, test_data.targets
run_sklearn	250–355	args, data, logger	Dict[str, List[float]]	debug, get_data, get_task_names, split_data, get_features_generator, args.as_dict, ValueError, save_smiles_splits, tqdm,

				single_task_sklearn, multi_task_sklearn, info, data.num_tasks, RandomForestRegressor
sklearn_train	358–363		None	cross_validate, SklearnTrainArgs().parse_args, SklearnTrainArgs

Models/polymer_periodic_graph/Polymer_Model/chemprop/spectra_utils.py

函数/类	行号	参数	返回	静态调用（前 14 项）
sid_loss	9–39	model_spectra, target_spectra, mask, threshold	torch.tensor	torch.zeros_like, torch.ones_like, torch.where, torch.sum, torch.div, loss.to, torch.full, torch.mul, torch.log
sid_metric	42–83	model_spectra, target_spectra, threshold, batch_size	float	trange, np.mean, len, np.array, np.sum, loss.tolist, losses.extend, np.log
wasserstein_loss	86–114	model_spectra, target_spectra, mask, threshold	torch.tensor	torch.zeros_like, torch.where, torch.sum, torch.div, torch.cumsum(target_spectra, axis=1).to, torch.cumsum(model_spectra, axis=1).to, torch.abs, loss.to, torch.full, torch.cumsum
wasserstein_metric	117–159	model_spectra, target_spectra, threshold, batch_size	float	trange, np.mean, len, np.array, np.sum, np.cumsum, np.abs, loss.tolist, losses.extend
normalize_spectra	162–208	spectra, phase_features, phase_mask, batch_size, excluded_sub_value, threshold	List[List[float]]	trange, np.array, len, np.sum, batch_spectra.astype, batch_spectra.tolist, normalized_spectra.extend, np.matmul(batch_phases, phase_mask).astype, np.matmul
roundrobin_sid	211–241	spectra, threshold	List[float]	range, len, np.isnan, np.zeros, np.sum, np.mean, ensemble_sids.append,

				np.concatenate, np.log
load_phase_mask	244–264	path	List[List[int]]	open, csv.reader, next, any, data.append, ValueError, int

Models/polymer_periodic_graph/Polymer_Model/chemprop/train/__init__.py

无可解析顶层函数/类，可能为脚本式逻辑或纯导入文件。

Models/polymer_periodic_graph/Polymer_Model/chemprop/train/cross_validate.py

函数/类	行号	参数	返回	静态调用（前 14 项）
cross_validate	21–181	args, train_func	Tuple[float, float]	timeit, create_logger, get_task_names, debug, makedirs, reset_featurization_parameters, set_explicit_h, set_reaction, get_data, validate_dataset_type, data.features_size, defaultdict, range, dict
chemprop_train	184–189		None	cross_validate, TrainArgs().parse_args, TrainArgs

Models/polymer_periodic_graph/Polymer_Model/chemprop/train/evaluate.py

函数/类	行号	参数	返回	静态调用（前 14 项）
evaluate_predictions	11–80	preds, targets, num_tasks, metrics, dataset_type, logger	Dict[str, List[float]]	defaultdict, dict, get_metric_func, len, range, metric_to_func.items, results[metric].append, metric_func, float, valid_preds[i].append, valid_targets[i].append, all, info, list
evaluate	83–118	model, data_loader, num_tasks, metrics, dataset_type, scaler, logger	Dict[str, List[float]]	predict, evaluate_predictions

Models/polymer_periodic_graph/Polymer_Model/chemprop/train/make_predictions.py

函数/类	行号	参数	返回	静态调用（前 14 项）
load_model	16–41	args, generator	未标注；含 return	print, load_args, update_prediction_args, load_checkpoint,

				load_scalers, list
load_data	43–83	args, smiles	未标注；含 return	print, range, MoleculeDataset, MoleculeDataLoader, get_data_from_smiles, get_data, len, all, sorted, full_to_valid_indices.keys
set_features	86–104	args, train_args	未标注；无显式 return	reset_featurization_parameters, set_explicit_h, set_reaction, set_extra_atom_fdim, set_extra_bond_fdim
predict_and_save	107–245	args, train_args, test_data, task_names, num_tasks, test_data_loader, full_data, full_to_valid_indices, models, scalers	未标注；含 return	print, enumerate, makedirs, avg_preds.tolist, np.zeros, tqdm, predict, np.array, len, full_to_valid_indices.get, zip, open, csv.DictWriter, writer.writeheader
make_predictions	249–278	args, smiles, model_objects	List[List[Optional[float]]]	timeit, set_features, load_data, predict_and_save, load_model, len
chemprop_predict	281–286		None	make_predictions, PredictArgs().parse_args, PredictArgs

Models/polymer_periodic_graph/Polymer_Model/chemprop/train/molecule_fingerprint.py

函数/类	行号	参数	返回	静态调用（前 14 项）
molecule_fingerprint	16–147	args, smiles	List[List[Optional[float]]]	timeit, print, load_args, update_prediction_args, reset_featurization_parameters, set_explicit_h, set_reaction, range, MoleculeDataset, MoleculeDataLoader, np.zeros, enumerate, makedirs, get_data_from_smiles
model_fingerprint	149–181	model, data_loader, fingerprint_type, disable_progress_bar	List[List[float]]	model.eval, tqdm, batch_fp.data.cpu().tolist, fingerprints.extend, batch.batch_graph

				, batch.features, batch.atom_descriptors, batch.atom_features, batch.bond_features, torch.no_grad, model.fingerprint, batch_fp.data.cpu
chemprop_fingerprint	183–188		None	molecule_fingerprint, FingerprintArgs().parse_args, FingerprintArgs

Models/polymer_periodic_graph/Polymer_Model/chemprop/train/predict.py

函数/类	行号	参数	返回	静态调用（前 14 项）
predict	10–48	model, data_loader, disable_progress_bar, scaler	List[List[float]]	model.eval, tqdm, batch_preds.data.cpu().numpy, batch_preds.tolist, preds.extend, batch.batch_graph, batch.features, batch.atom_descriptors, batch.atom_features, batch.bond_features, torch.no_grad, model, scaler.inverse_transform, batch_preds.data.cpu

Models/polymer_periodic_graph/Polymer_Model/chemprop/train/run_training.py

函数/类	行号	参数	返回	静态调用（前 14 项）
run_training	26–364	args, data, logger	Dict[str, List[float]]	torch.manual_seed, debug, len, get_loss_func, MoleculeDataLoader, range, (sum_test_preds / args.ensemble_size).tolist, evaluate_predictions, ensemble_scores.items, get_data, get_class_sizes, enumerate, save_smiles_splits, train_data.normalize_features

Models/polymer_periodic_graph/Polymer_Model/chemprop/train/train.py

函数/类	行号	参数	返回	静态调用（前 14 项）

train	17–113	model, data_loader, loss_func, optimizer, scheduler, args, n_iter, logger, writer	int	model.train, tqdm, torch.tensor, torch.Tensor(data_weights_batch).unsqueeze, model.zero_grad, model, mask.to, targets.to, target_weights.to, data_weights.to, loss.item, loss.backward, optimizer.step, isinstance

Models/polymer_periodic_graph/Polymer_Model/chemprop/utils.py

函数/类	行号	参数	返回	静态调用（前 14 项）
makedirs	29–42	path, isfile	None	os.path.dirname, os.makedirs
save_checkpoint	45–87	path, model, scaler, features_scaler, atom_descriptor_scaler, bond_feature_scaler, args	None	torch.save, Namespace, model.state_dict, args.as_dict
load_checkpoint	90–147	path, device, logger	MoleculeModel	torch.load, TrainArgs, args.from_dict, MoleculeModel, model.state_dict, loaded_state_dict.keys, model_state_dict.update, model.load_state_dict, model.to, vars, re.match, debug, loaded_param_name.replace, info
overwrite_state_dict	150–179	loaded_param_name, model_param_name, loaded_state_dict, model_state_dict, logger	collections.OrderedDict	debug
load_frzn_model	181–267	model, path, current_args, cuda, logger	MoleculeModel	torch.load, model.state_dict, model.load_state_dict, debug, overwrite_state_dict, zip, Exception, range, 'Number of molecules in checkpoint_frzn ({}) must match current model ({}) OR equal to 1.'.format, 'Number of molecules in checkpoint_frzn

				({}) must be equal to 1 for freeze_first_only to be used.'.format, str
load_scalers	269–299	path	Tuple[StandardScaler, StandardScaler, StandardScaler, StandardSca	torch.load, StandardScaler, state.keys
load_args	302–312	path	TrainArgs	TrainArgs, args.from_dict, vars, torch.load
load_task_names	315–322	path	List[str]	load_args
get_loss_func	325–351	args	nn.Module	ValueError, nn.BCEWithLogitsLoss, nn.MSELoss, nn.CrossEntropyLoss
prc_auc	354–363	targets, preds	float	precision_recall_curve, auc
bce	366–378	targets, preds	float	nn.BCELoss, bce_func(target=torch.Tensor(targets), input=torch.Tensor(preds)).item, bce_func, torch.Tensor
rmse	381–389	targets, preds	float	math.sqrt, mean_squared_error
mse	392–400	targets, preds	float	mean_squared_error
accuracy	403–419	targets, preds, threshold	float	accuracy_score, type, p.index, max
get_metric_func	422–474	metric	Callable[[Union[List[int], List[float]], List[float]], float]	ValueError
build_optimizer	477–487	model, args	Optimizer	Adam, model.parameters
build_lr_scheduler	490–508	optimizer, args, total_epochs	_LRScheduler	NoamLR
create_logger	511–554	name, save_dir, quiet	logging.Logger	logging.getLogger, logger.setLevel, logging.StreamHandler, logger.addHandler, ch.setLevel, makedirs, logging.FileHandler, fh_v.setLevel, fh_q.setLevel, os.path.join
timeit	557–583	logger_name	Callable[[Callable], Callable]	wraps, time, func, timedelta, info, round, logging.getLogger
save_smiles_splits	586–695	data_path, save_dir, task_names, features_path, train_data, val_data, test_data, logger,	None	makedirs, isinstance, preprocess_smiles_columns, open, csv.DictReader, enumerate, get_task_names,

		smiles_columns		tqdm, tuple, csv.writer, dataset.smiles, writer.writerow, dataset.targets, dataset.features
update_prediction_args	698–774	predict_args, train_args, missing_to_defaults, validate_feature_sources	None	vars(train_args).items, TrainArgs().parse_args, vars(default_train_args).items, ValueError, vars, hasattr, setattr, TrainArgs, str, override_defaults.get

Models/polymer_periodic_graph/Polymer_Model/predict.py

无可解析顶层函数/类，可能为脚本式逻辑或纯导入文件。

Models/polymer_periodic_graph/Polymer_Model/setup.py

无可解析顶层函数/类，可能为脚本式逻辑或纯导入文件。

Models/polymer_periodic_graph/Polymer_Model/train.py

无可解析顶层函数/类，可能为脚本式逻辑或纯导入文件。

scripts/normalize_legacy_full_dataset_results.py

函数/类	行号	参数	返回	静态调用（前 14 项）
parse_args	51–55		argparse.Namespace	argparse.ArgumentParser, parser.add_argument, parser.parse_args
normalized_table	58–102	source	pd.DataFrame	pd.DataFrame, (result[['dataset_size', 'rmse_mean', 'rmse_std']] < 0).any().any, result.duplicated(identifiers).any, result.sort_values(identifiers).reset_index, list, ValueError, set, MODEL_NAMES.keys, PROPERTY_NAMES.keys, ARCHITECTURES.keys, pd.to_numeric, source['Quantity'].map, source['Model'].map, source['PolymerType'].map
main	105–111		None	parse_args, pd.read_csv, normalized_table, args.output.parent.

				mkdir, result.to_csv, print, len

scripts/summarize_scaling_results.py

函数/类	行号	参数	返回	静态调用（前 14 项）
json_default	17–20	value	未标注；含 return	hasattr, TypeError, value.item, type
parse_args	23–33		argparse.Namespace	argparse.ArgumentParser, parser.add_argument, parser.parse_args
parse_train_and_fold	36–56	metrics_path	tuple[int | None, int | None]	enumerate, part.startswith, int, parts[index + 1].startswith, len, part.split, parts[index + 1].split
experiment_name	59–64	result_root	str	result_root.resolve, parts.index, str, Path
experiment_metadata	67–96	experiment	dict[str, str | None]	Path, len, file_stem.startswith
model_name	99–104	result_root	str	result_root.resolve
portable_path	107–112	path	str	str, path.resolve().relative_to, path.resolve
rows_from_metrics	115–143	metrics_path, result_root	list[dict]	parse_train_and_fold, experiment_name, experiment_metadata, data['metrics'].items, open, json.load, rows.append, data.get, metrics.get, portable_path
test_size_for_chemprop	146–150	test_scores_path	int | None	max, test_full.exists, sum, open
rows_from_chemprop_scores	153–183	test_scores_path, result_root	list[dict]	parse_train_and_fold, experiment_name, experiment_metadata, test_size_for_chemprop, data.items, open, json.load, rows.append, metric_name.lower, isinstance, model_name, float, portable_path
main	186–236		None	parse_args, args.out_dir.resolve, out_dir.mkdir, pd.DataFrame, detailed.sort_value

				s, detailed.to_csv, detailed.groupby(['experiment', 'dataset', 'quantity', 'polymer_type', 'model', 'train_size', 'split'], as_index=False).agg(rmse_mean=('rmse', 'mean'), rmse_std=('rmse', lambda x: float(np.std(x, ddof=0))), mae_mean=('mae', 'mean'), mae_std=('mae', lambda x: float(np.std(x, ddof=0))), mse_mean=('mse', 'mean'), n_folds=('fold', 'nunique'), n_samples_mean=('n_samples', 'mean')).sort_values, aggregate.to_csv, print, root.resolve, sorted, ValueError, open, json.dump

tests/test_dataset_construction.py

函数/类	行号	参数	返回	静态调用（前 14 项）
run_cli	37–43	script	subprocess.CompletedProcess[str]	subprocess.run, str
tree_bytes	46–51	root	dict[str, bytes]	str, path.read_bytes, path.relative_to, sorted, path.is_file, root.rglob
ConversionUnitTests	54–135	class	类实例	—
ConversionUnitTests.test_rdkit_descriptor_cache_reuses_exact_psmiles	55–70	self	None	tempfile.TemporaryDirectory, pd.testing.assert_frame_equal, self.assertTrue, Path, DescriptorCache, cache_path.is_file, patch, cache.descriptor_frame, self.assertEqual
ConversionUnitTests.test_task4_component_parsing_and_weighted_prediction	72–77	self	None	self.assertEqual, self.assertAlmostEqual, parse_wpsmiles, weighted_prediction, self.assertRaises
ConversionUnitT	79–87	self	None	pd.DataFrame,

ests.test_task4_shared_folds_are_deterministic_and_grouped				assign_folds, np.testing.assert_array_equal, self.assertEqual, self.assertTrue, set, range
ConversionUnitTests.test_homopolymer_round_trip	89–96	self	None	PSMILES_to_wPSMILES, self.assertEqual, wPSMILES_to_PSMILES_homopolymer_simple
ConversionUnitTests.test_copolymer_conversions	98–108	self	None	construct_alternating_wPSMILES, self.assertIn, wPSMILES_to_PSMILES_alternating, self.assertEqual, construct_random_wPSMILES, converted.count
ConversionUnitTests.test_split_indices_are_deterministic_and_disjoint	110–121	self	None	split_indices, zip, self.assertTrue, self.assertEqual, np.testing.assert_array_equal, set, train.isdisjoint, val.isdisjoint, len, self.assertRaisesRegex, values.tolist
ConversionUnitTests.test_invalid_psmiles_are_rejected_before_descriptor_generation	123–135	self	None	pd.DataFrame, RDLogger.DisableLog, RDLogger.EnableLog, redirect_stdout, io.StringIO, self.assertRaisesRegex, build_rdkit_frame
ConstructionIntegrationTests	138–486	class	类实例	—
ConstructionIntegrationTests.test_vipea_processor_creates_vipea_output_directories	139–187	self	None	pd.DataFrame, tempfile.TemporaryDirectory, self.assertFalse, Path, patch, process_vipea_data, self.assertTrue, (datasets_root / 'PSMILES' / 'Coley_2022').exists, (datasets_root / 'wPSMILES' / 'Coley_2022').exists, (datasets_root / 'PSMILES' / 'Vipea' / quantity / f'alternating_{quantity}.csv').is_file, (datasets_root / 'wPSMILES' / 'Vipea' / quantity / f'{architecture}_{quantity}.csv').is_file

ConstructionIntegrationTests.test_converter_is_non_mutating_and_strict_missing_fails	189–224	self	None	tempfile.TemporaryDirectory, Path, input_root.mkdir, source.write_text, source.read_bytes, run_cli, self.assertEqual, pd.read_csv, self.assertTrue, self.assertNotEqual, self.assertIn, str, len, converted['smiles'].str.contains('\\|', regex=True).all
ConstructionIntegrationTests.test_tiny_pipeline_alignment_and_determinism	226–426	self	None	tempfile.TemporaryDirectory, Path, psmiles_path.parent.mkdir, pd.DataFrame, canonical.to_csv, psmiles_path.read_bytes, run_cli, self.assertEqual, np.add.outer, embedding_path.parent.mkdir, embedding_frame.to_csv, json.loads, self.assertNotIn, range
ConstructionIntegrationTests.test_scaling_builder_rejects_misaligned_representations	428–473	self	None	tempfile.TemporaryDirectory, Path, pd.DataFrame, canonical.copy, weighted['smiles'].map, run_cli, self.assertNotEqual, self.assertIn, path.parent.mkdir, frame.to_csv, str, np.arange, range
ConstructionIntegrationTests.test_cli_help_works_outside_repository	475–486	self	None	tempfile.TemporaryDirectory, Path, self.subTest, run_cli, self.assertEqual

tests/test_fresh_clone.py

函数/类	行号	参数	返回	静态调用（前 14 项）
run	24–31		subprocess.CompletedProcess[str]	subprocess.run
FreshCloneTests	34–275	class	类实例	—
FreshCloneTests.setUpClass	36–37	cls	None	json.loads, MANIFEST_PATH.read_text

FreshCloneTests.test_required_sources_are_tracked_and_present_in_git_archive	39–55	self	None	set, run('git', 'ls-files').stdout.splitlines, subprocess.run, tarfile.open, handle.getnames, self.subTest, self.assertIn, self.assertGreater, io.BytesIO, run, (REPO_ROOT / relative).stat
FreshCloneTests.test_required_source_schemas	57–81	self	None	self.subTest, self.assertTrue, path.open, json.loads, set, next, source.get, found.update, csv.reader
FreshCloneTests.test_generated_artifacts_are_ignored	83–87	self	None	self.subTest, run, self.assertEqual
FreshCloneTests.test_task3_complexity_metadata_is_complete_and_consistent	89–107	self	None	pd.read_csv, self.assertFalse, self.assertTrue, pd.testing.assert_series_equal, self.assertEqual, frame.isna().any().any, frame['legacy_test_fold'].isin(range(1, 6)).all, frame['wpsmiles'].str.split('|').str[0].str.count, fragments.reset_index, frame['n_distinct_monomers'].reset_index, frame['n_distinct_monomers'].between, len, task3['n_distinct_monomers'].isin(range(5, 11)).all, frame.isna().any
FreshCloneTests.test_task4_legacy_training_source_is_complete	109–117	self	None	pd.read_csv, self.assertEqual, self.assertFalse, list, len, frame.isna().any().any, int, frame['SMILES'].duplicated().sum, frame.isna().any, frame['SMILES'].duplicated, frame.isna
FreshCloneTests.test_tracked_construction_code_has_no_machine_specific_paths	119–132	self	None	run('git', 'ls-files', 'Datasets/Dataset_construction_scripts/*.py', 'Datasets/Dataset

				_construction_scripts/*.sh').stdout.splitlines, re.compile, self.subTest, (REPO_ROOT / relative).read_text, self.assertIsNone, run, absolute_home.search, retired_name.search
FreshCloneTests.test_scaling_driver_matches_manifest	134–164	self	None	run, result.stdout.splitlines, self.assertEqual, shlex.split, next, actual.append, line.startswith, line.removeprefix('DRY RUN:').strip, args.index, value, line.removeprefix, range, args[index].startswith, int
FreshCloneTests.test_published_scaling_packages_match_manifest_and_are_trackable	166–197	self	None	sorted, set, self.assertEqual, root.rglob, run('git', 'ls-files', settings['root']).stdout.splitlines, json.loads, actual.add, tuple, manifest_path.read_text, str, path.relative_to, self.subTest, self.assertIn, run
FreshCloneTests.test_real_condition_materializes_published_indices	199–275	self	None	json.loads, hashlib.sha256(canonical_path.read_bytes()).hexdigest, self.assertEqual, (package_base / 'manifest.json').read_text, canonical_path.is_file, self.skipTest, tempfile.TemporaryDirectory, run, enumerate, hashlib.sha256, Path, str, (output_base / 'manifest.json').read_text, expected_arrays.items

6. 模型训练/评估关键文件详细索引

Datasets/Dataset_construction_scripts/create_scaling_splits.py

repo_root (L32–33) | args: | returns: Path |

parse_args (L36–95) | args: | returns: argparse.Namespace |

target_columns (L98–101) | args: df, representation | returns: list[str] |

require_single_target (L104–110) | args: df, representation | returns: str |

dataset_path (L113–114) | args: root, representation, dataset, prop, filename | returns: Path |

load_and_validate_inputs (L117–164) | args: args | returns: tuple[dict[str, pd.DataFrame], str] |

build_rdkit_frame (L167–188) | args: psmiles, target_col, cache_path | returns: pd.DataFrame |

write_csv (L191–195) | args: df, path, overwrite | returns: None |

split_indices (L198–214) | args: n_rows, fold, seed, val_frac, test_frac | returns: tuple[np.ndarray, np.ndarray, np.ndarray] |

indices_frame (L217–227) | args: indices, split, psmiles, target_col | returns: pd.DataFrame |

output_base (L230–231) | args: args | returns: Path |

load_published_indices (L234–297) | args: args, psmiles_path, n_rows, target_col | returns: tuple[dict, list[tuple[np.ndarray, np.ndarray, np.ndarray]]] | None |

Models/RDKit_RF/predict_rf_ensemble.py

calc_all_descriptors (L15–19) | args: psmi, missing | returns: 未标注；含 return |

sanitize (L22–37) | args: X | returns: pd.DataFrame |

convert_file (L40–51) | args: csv_path | returns: 未标注；含 return |

align_columns (L54–59) | args: X, train_columns | returns: pd.DataFrame |

find_fold_models (L63–73) | args: root | returns: 未标注；含 return |

load_train_columns (L76–93) | args: fold_dir | returns: 未标注；含 return |

main (L96–178) | args: | returns: 未标注；无显式 return |

Models/RDKit_RF/train_rf.py

sanitize_features (L27–49) | args: X | returns: tuple[pd.DataFrame, list[str]] | Remove problematic columns and handle inf/nan values.

load_and_prepare_data (L52–95) | args: data_path | returns: 未标注；含 return | Load RDKit descriptor dataset and separate features from target.

train_random_forest (L98–117) | args: X_train, y_train, random_state | returns: 未标注；含 return | Train a Random Forest model with preprocessing pipeline.

evaluate_model (L120–132) | args: model, X, y, split_name | returns: 未标注；含 return | Evaluate model and return metrics.

main (L135–281) | args: | returns: 未标注；无显式 return |

Models/RDKit_RF/train_rf_scaling.py

json_default (L21–24) | args: value | returns: 未标注；含 return |

parse_args (L27–34) | args: | returns: argparse.Namespace |

load_row_indices (L37–43) | args: path, expected_rows | returns: np.ndarray |

train_random_forest (L46–62) | args: x_train, y_train, random_state, n_estimators | returns: Pipeline |

evaluate (L65–78) | args: model, x, y, split | returns: tuple[dict, np.ndarray] |

write_predictions (L81–89) | args: path, original_indices, y_true, y_pred | returns: None |

align_columns (L92–96) | args: x, feature_cols | returns: pd.DataFrame |

train_one_split (L99–148) | args: split_dir, output_dir, seed, n_estimators | returns: dict |

train_size_from_dir (L151–152) | args: path | returns: int |

write_summary (L155–179) | args: rows, save_root | returns: None |

main (L182–222) | args: | returns: None |

Models/polyBERT/pBERT_predict_ensemble.py

FeedForwardNet.__init__ (L34–53) | args: self, input_dim, cfg | returns: 未标注；无显式 return |

FeedForwardNet.forward (L55–56) | args: self, x | returns: 未标注；含 return |

XOnlyDataset.__init__ (L60–61) | args: self, X | returns: 未标注；无显式 return |

XOnlyDataset.__len__ (L63–64) | args: self | returns: 未标注；含 return |

XOnlyDataset.__getitem__ (L66–67) | args: self, idx | returns: 未标注；含 return |

predict_one_checkpoint (L71–115) | args: ckpt_path, X_raw, batch_size, device | returns: np.ndarray |

load_csv_as_array (L118–129) | args: csv_path, x_cols, y_col, skip_header | returns: Tuple[np.ndarray, Optional[np.ndarray]] |

main (L132–206) | args: | returns: 未标注；无显式 return |

Models/polyBERT/train_pBERT.py

make_out_dir_from_dataset (L44–55) | args: dataset_path, anchor | returns: 未标注；含 return |

ArrayDataset.__init__ (L59–65) | args: self, X, y | returns: 未标注；无显式 return |

ArrayDataset.set_norm (L67–69) | args: self, mean, std | returns: 未标注；无显式 return |

ArrayDataset.__len__ (L71–71) | args: self | returns: 未标注；含 return |

ArrayDataset.__getitem__ (L73–78) | args: self, idx | returns: 未标注；含 return |

FeedForwardNet.__init__ (L82–105) | args: self, input_dim, cfg | returns: 未标注；无显式 return |

FeedForwardNet.forward (L107–108) | args: self, x | returns: 未标注；含 return |

evaluate (L111–138) | args: model, loader, criterion, device, is_classification | returns: 未标注；含 return |

predict_with_indices (L141–155) | args: model, loader, device, is_classification | returns: 未标注；含 return |

train_one_epoch (L158–171) | args: model, loader, criterion, optimizer, device, scheduler | returns: 未标注；含 return |

Models/polyBERT/train_pBERT_scaling.py

json_default (L22–25) | args: value | returns: 未标注；含 return |

parse_args (L28–37) | args: | returns: argparse.Namespace |

load_polybert_csv (L40–49) | args: path | returns: tuple[np.ndarray, np.ndarray] |

load_row_indices (L52–58) | args: path, expected_rows | returns: np.ndarray |

metrics_from_arrays (L61–71) | args: y_true, y_pred, split | returns: dict[str, float | int | str] |

predict_split (L74–77) | args: model, dataset, batch_size, device | returns: tuple[np.ndarray, np.ndarray, np.ndarray] |

write_predictions (L80–88) | args: path, original_indices, y_true, y_pred | returns: None |

train_one_split (L91–206) | args: split_dir, output_dir, cfg, seed, device, quiet | returns: dict |

train_size_from_dir (L209–210) | args: path | returns: int |

write_summary (L213–239) | args: rows, save_root | returns: None |

main (L242–291) | args: | returns: None |

Models/polymer_periodic_graph/Polymer_Model/chemprop/train/__init__.py

Models/polymer_periodic_graph/Polymer_Model/chemprop/train/cross_validate.py

cross_validate (L21–181) | args: args, train_func | returns: Tuple[float, float] | Runs k-fold cross-validation.

chemprop_train (L184–189) | args: | returns: None | Parses Chemprop training arguments and trains (cross-validates) a Chemprop model.

Models/polymer_periodic_graph/Polymer_Model/chemprop/train/evaluate.py

evaluate_predictions (L11–80) | args: preds, targets, num_tasks, metrics, dataset_type, logger | returns: Dict[str, List[float]] | Evaluates predictions using a metric function after filtering out invalid targets.

evaluate (L83–118) | args: model, data_loader, num_tasks, metrics, dataset_type, scaler, logger | returns: Dict[str, List[float]] | Evaluates an ensemble of models on a dataset by making predictions and then evaluating the predictions.

Models/polymer_periodic_graph/Polymer_Model/chemprop/train/make_predictions.py

load_model (L16–41) | args: args, generator | returns: 未标注；含 return | Function to load a model or ensemble of models from file. If generator is True, a generator of the respective model and scaler

load_data (L43–83) | args: args, smiles | returns: 未标注；含 return | Function to load data from a list of smiles or a file.

set_features (L86–104) | args: args, train_args | returns: 未标注；无显式 return | Function to set extra options.

predict_and_save (L107–245) | args: args, train_args, test_data, task_names, num_tasks, test_data_loader, full_data, full_to_valid_indices, models, scalers | returns: 未标注；含 return | Function to predict with a model and save the predictions to file.

make_predictions (L249–278) | args: args, smiles, model_objects | returns: List[List[Optional[float]]] | Loads data and a trained model and uses the model to make predictions on the data.

chemprop_predict (L281–286) | args: | returns: None | Parses Chemprop predicting arguments and runs prediction using a trained Chemprop model.

Models/polymer_periodic_graph/Polymer_Model/chemprop/train/molecule_fingerprint.py

molecule_fingerprint (L16–147) | args: args, smiles | returns: List[List[Optional[float]]] | Loads data and a trained model and uses the model to encode fingerprint vectors for the data.

model_fingerprint (L149–181) | args: model, data_loader, fingerprint_type, disable_progress_bar | returns: List[List[float]] | Encodes the provided molecules into the latent fingerprint vectors, according to the provided model.

chemprop_fingerprint (L183–188) | args: | returns: None | Parses Chemprop predicting arguments and returns the latent representation vectors for

Models/polymer_periodic_graph/Polymer_Model/chemprop/train/predict.py

predict (L10–48) | args: model, data_loader, disable_progress_bar, scaler | returns: List[List[float]] | Makes predictions on a dataset using an ensemble of models.

Models/polymer_periodic_graph/Polymer_Model/chemprop/train/run_training.py

run_training (L26–364) | args: args, data, logger | returns: Dict[str, List[float]] | Loads data, trains a Chemprop model, and returns test scores for the model checkpoint with the highest validation score.

Models/polymer_periodic_graph/Polymer_Model/chemprop/train/train.py

train (L17–113) | args: model, data_loader, loss_func, optimizer, scheduler, args, n_iter, logger, writer | returns: int | Trains a model for an epoch.

scripts/normalize_legacy_full_dataset_results.py

parse_args (L51–55) | args: | returns: argparse.Namespace |

normalized_table (L58–102) | args: source | returns: pd.DataFrame |

main (L105–111) | args: | returns: None |

scripts/summarize_scaling_results.py

json_default (L17–20) | args: value | returns: 未标注；含 return |

parse_args (L23–33) | args: | returns: argparse.Namespace |

parse_train_and_fold (L36–56) | args: metrics_path | returns: tuple[int | None, int | None] |

experiment_name (L59–64) | args: result_root | returns: str |

experiment_metadata (L67–96) | args: experiment | returns: dict[str, str | None] | Extract dataset metadata from a scaling-results experiment path.

model_name (L99–104) | args: result_root | returns: str |

portable_path (L107–112) | args: path | returns: str | Return a repository-relative path when the result lives in this checkout.

rows_from_metrics (L115–143) | args: metrics_path, result_root | returns: list[dict] |

test_size_for_chemprop (L146–150) | args: test_scores_path | returns: int | None |

rows_from_chemprop_scores (L153–183) | args: test_scores_path, result_root | returns: list[dict] |

main (L186–236) | args: | returns: None |

7. 多层级 3D 模型接入：源码映射与改造接口

当前模块	现有职责	建议改造/新增	科学意义
featurization.py:MolGraph	2D atom/bond graph	新增 3D 坐标、非键空间邻居、FG membership、unit membership	引入链构象与非键相互作用
featurization.py:BatchMolGraph	批图索引与 padding	新增 fg_scope、unit_scope、atom2fg、atom2unit、3D batch index	支持不同 DP、变长单链
mpn.py:MPNEncoder	原子/键消息传递	新增 3D GNN + Ewald-like long-range branch	局域几何与长程耦合
新增 **FG encoder**	当前未见独立 FG 层	FG graph、FG pair、刚柔连接特征	官能团间物理相互作用与可解释性
新增 **Unit encoder**	当前无独立重复单元层	unit graph、unit pair、sequence	保留聚合物链重复单元
新增 **Inter-level**	当前原子图主路径	atom↔unit 向增强，再与 FG 增强表示融合	避免过早三层交互的冗余
model.py:MoleculeModel	graph readout + FFN	multi-level fusion + chain readout	建立结构—性质关联
data/utils.py / train	读取 CSV 与监督训练	增加 3D 结构文件索引、缺失构象处理、共享 split	公平基准和避免数据泄漏

重要限制：PSMILES 本身不提供真实单链 3D 坐标。应明确构象生成/MD 来源、DP、立构、随机序列和模拟条件；不要把基于单体生成的 3D 构象当成实验测量的链结构。

8. 静态审查结果与验证优先级

本技术文档完成了全量 Python AST 定义与调用清单，但不能代替运行时 profiler。优先动态验证：①数据字段/分割一致性；② featurization 真实 F_atom/F_bond；③ atom_messages 配置；④ periodic boundary 图连接细节；⑤模型输出维度/失；⑥ checkpoint 入；⑦外部 Chemprop 版本与权重处理。